Planning

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Chapter 3

文件索引

3.1 文件列表

里列出了所有文档化的文件，并附要明:

/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/**define**↩
Color.h . . . . . ??
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/**geometry**↩
_m.h . . . . . ??
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/**global**↩
Planning_m.h . . . . . ??
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/**interpolate**↩
_m.h . . . . . ??
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/**local**↩
Planning_m.hpp . . . . . ??
/home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/**planning.**↩
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Chapter 4

Planning

共 9 + 16 + 2 + 48 = 75 个子模

4.1 geometry 功能模

是一个算欧式距离的namespace。

- struct crossInput
- struct crossOutput
- struct crossParam
- struct inOrOutPolygonInput
- struct inOrOutPolygonOutput
- struct inOrOutPolygonParam
- struct isInSimpleInput
- struct isInSimpleOutput
- struct isInSimpleParam
- struct isLineIntersectionInput
- struct isLineIntersectionOutput
- struct isLineIntersectionParam
- struct isOnlineInput
- struct isOnlineOutput
- struct isOnlineParam
- struct isPointInPolygonInput
- struct isPointInPolygonOutput
- struct isPointInPolygonParam
- struct isPolygonsIntersectionInput
- struct isPolygonsIntersectionOutput
- struct isPolygonsIntersectionParam
- struct point_t
- struct pointToLineInput
- struct pointToLineOutput
- struct pointToLineParam
- struct pointToPolygonInput
- struct pointToPolygonOutput
- struct pointToPolygonParam

函数

9 个子模

- void pointToPolygon (const pointToPolygonParam &param, const pointToPolygonInput &input, pointToPolygonOutput &output)

  算点到凸多形的距离
- void cross (const crossParam &param, const crossInput &input, crossOutput &output)

  点乘
- void pointToLine (const pointToLineParam &param, const pointToLineInput &input, pointToLineOutput &output)

  算点到段的距离
- void isOnline (const isOnlineParam &param, const isOnlineInput &input, isOnlineOutput &output)

  判断点是否在段上
- void isInSimple (const isInSimpleParam &param, const isInSimpleInput &input, isInSimpleOutput &output)

  判断点是否在横平直的多形内部
- void isLineIntersection (const isLineIntersectionParam &param, const isLineIntersectionInput &input, isLineIntersectionOutput &output)

  判断两段是否相交
- void isPolygonsIntersection (const isPolygonsIntersectionParam &param, const isPolygonsIntersectionInput &input, isPolygonsIntersectionOutput &output)

  判断两多形是否相交
- void isPointInPolygon (const isPointInPolygonParam &param, const isPointInPolygonInput &input, isPointInPolygonOutput &output)

  判断点是否在多形内部
- void inOrOutPolygon (const inOrOutPolygonParam &param, const inOrOutPolygonInput &input, inOrOutPolygonOutput &output)

  判断点是否在多形内部

4.1.1 cross()

```
void geometry::cross (
            const crossParam & param,
            const crossInput & input,
            crossOutput & output )
```

点乘

参数

[IN]	param 无
[IN]	input 向量原点，
[OUT]	output 点到凸多形的距离 @cn_name: 点乘 @granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

4.1.2 inOrOutPolygon()

```
void geometry::inOrOutPolygon (
            const inOrOutPolygonParam & param,
            const inOrOutPolygonInput & input,
            inOrOutPolygonOutput & output )
```

判断点是否在多形内部

参数

[IN]	param 无
[IN]	input 多形点数、多形点X，多形点Y，点X，点Y
[OUT]	output 判断果 @cn_name: 判断点是否在多形内部 @granularity: atomic //函数件的粒度，atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

4.1.3 isInSimple()

```
void geometry::isInSimple (
            const isInSimpleParam & param,
            const isInSimpleInput & input,
            isInSimpleOutput & output )
```

判断点是否在横平直的多形内部

参数

[IN]	param 无
[IN]	input 多形点，多形点数，点
[OUT]	output 判断果 @cn_name: 判断点是否在横平直的多形内部 @granularity: composite //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

4.1.4 isLineIntersection()

```
void geometry::isLineIntersection (
            const isLineIntersectionParam & param,
            const isLineIntersectionInput & input,
            isLineIntersectionOutput & output )
```

判断两段是否相交

参数

[IN]	param 无
[IN]	input 两段起点、点
[OUT]	output 判断果 @cn_name: 判断两段是否相交 @granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

4.1.5 isOnline()

```
void geometry::isOnline (
            const isOnlineParam & param,
            const isOnlineInput & input,
            isOnlineOutput & output )
```

判断点是否在段上

参数

[IN]	param 无
[IN]	input 段起点，段点，点
[OUT]	output 判断果 @cn_name: 判断点是否在段上 @granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

4.1.6 isPointInPolygon()

```
void geometry::isPointInPolygon (
            const isPointInPolygonParam & param,
            const isPointInPolygonInput & input,
            isPointInPolygonOutput & output )
```

判断点是否在多形内部

参数

[IN]	param 无
[IN]	input 多形点、点
[OUT]	output 判断果 @cn_name: 判断点是否在多形内部 @granularity: composite //函数件的粒度，atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

4.1.7 isPolygonsIntersection()

```
void geometry::isPolygonsIntersection (
            const isPolygonsIntersectionParam & param,
            const isPolygonsIntersectionInput & input,
            isPolygonsIntersectionOutput & output )
```

判断两多形是否相交

参数

[IN]	param 无
[IN]	input 两多形点
[OUT]	output 判断果 @cn_name: 判断两多形是否相交 @granularity: composite //函数件的粒度，atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

4.1.8 pointToLine()

```
void geometry::pointToLine (
            const pointToLineParam & param,
            const pointToLineInput & input,
            pointToLineOutput & output )
```

算点到段的距离

参数

[IN]	param 无
[IN]	input 段起点，段点，点
[OUT]	output 点到段的距离 @cn_name: 算点到段的距离 @granularity: atomic //函数件的粒度，atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

4.1.9 pointToPolygon()

```
void geometry::pointToPolygon (
            const pointToPolygonParam & param,
            const pointToPolygonInput & input,
            pointToPolygonOutput & output )
```

算点到凸多形的距离

参数

[IN]	param 无
[IN]	input 多形点向量数，点
[OUT]	output 点到凸多形的距离 @cn_name: 算点到凸多形的距离 @granularity: composite //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

4.2 GolbalPlanning 功能模

是一个名GolbalPlanning的namespace。

- class Astar
- struct AstarCalcFInput
- struct AstarCalcFOutput
- struct AstarCalcFParam
- struct AstarCalcGInput
- struct AstarCalcGOutput
- struct AstarCalcGParam
- struct AstarCalcHInput
- struct AstarCalcHOutput
- struct AstarCalcHParam
- struct AstarFindLaneInput
- struct AstarFindLaneOutput
- struct AstarFindLaneParam
- struct AstarFindLaneRInput
- struct AstarFindLaneROutput
- struct AstarFindLaneRParam
- struct AstarFindPathInput
- struct AstarFindPathOutput
- struct AstarFindPathParam
- struct AstarGetLeastFInput
- struct AstarGetLeastFOutput
- struct AstarGetLeastFParam
- struct AstarGetNextRoadInput
- struct AstarGetNextRoadOutput
- struct AstarGetNextRoadParam
- struct AstarGetPathInput
- struct AstarGetPathOutput
- struct AstarGetPathParam
- struct AstarInitLinkInput
- struct AstarInitLinkOutput
- struct AstarInitLinkParam
- struct AstarInitRoadInput
- struct AstarInitRoadOutput
- struct AstarInitRoadParam
- struct AstarMapToAstarInput
- struct AstarMapToAstarOutput
- struct AstarMapToAstarParam
- struct AstarModuleSelfCheckInput
- struct AstarModuleSelfCheckOutput
- struct AstarModuleSelfCheckParam
- struct AstarModuleSelfCheckPrintInput
- struct AstarModuleSelfCheckPrintOutput
- struct AstarModuleSelfCheckPrintParam
- struct AstarRecursiveFindLaneInput
- struct AstarRecursiveFindLaneOutput
- struct AstarRecursiveFindLaneParam
- struct AstarResetInput
- struct AstarResetOutput
- struct AstarResetParam
- class cmp
- class node
- struct road

函数

16 个子模

- void AstarMapToAstar (const AstarMapToAstarParam &param, const AstarMapToAstarInput &input, AstarMapToAstarOutput &output)

  使用*Map*构建*Astar*
- void AstarGetPath (const AstarGetPathParam &param, const AstarGetPathInput &input, AstarGetPathOutput &output)

  使用*Astar*算法取路径，包括理、后理等部分
- void AstarInitRoad (const AstarInitRoadParam &param, const AstarInitRoadInput &input, AstarInitRoadOutput &output)

  *Astar*道路初始化
- void AstarInitLink (const AstarInitLinkParam &param, const AstarInitLinkInput &input, AstarInitLinkOutput &output)

  *Astar*道路接系初始化
- void AstarReset (const AstarResetParam &param, const AstarResetInput &input, AstarResetOutput &output)

  在保留地的情况下，初始化*Astar*参数
- void AstarModuleSelfCheck (const AstarModuleSelfCheckParam &param, const AstarModuleSelfCheckInput &input, AstarModuleSelfCheckOutput &output)

  *Astar*道路*Id*
- void AstarModuleSelfCheckPrint (const AstarModuleSelfCheckPrintParam &param, const AstarModuleSelfCheckPrintInput &input, AstarModuleSelfCheckPrintOutput &output)

  *Astar*道路*Id*打印
- void AstarFindLane (const AstarFindLaneParam &param, const AstarFindLaneInput &input, AstarFindLaneOutput &output)

  根据*Astar*道路全局划果，得道全局划果
- void AstarFindLaneR (const AstarFindLaneRParam &param, const AstarFindLaneRInput &input, AstarFindLaneROutput &output)

  根据*Astar*道路全局划果，使用的方法得道全局划果
- void AstarRecursiveFindLane (const AstarRecursiveFindLaneParam &param, const AstarRecursiveFindLaneInput &input, AstarRecursiveFindLaneOutput &output)

  *Astar*得道全局划果主体函数
- void AstarFindPath (const AstarFindPathParam &param, const AstarFindPathInput &input, AstarFindPathOutput &output)

  *Astar*算法的主要部分
- void AstarCalcG (const AstarCalcGParam &param, const AstarCalcGInput &input, AstarCalcGOutput &output)

  *Astar*算法 算起点到当前点的已知度
- void AstarCalcH (const AstarCalcHParam &param, const AstarCalcHInput &input, AstarCalcHOutput &output)

  *Astar*算法 算当前点到点的估度
- void AstarCalcF (const AstarCalcFParam &param, const AstarCalcFInput &input, AstarCalcFOutput &output)

  *Astar*算法 算当前点估最路径度
- void AstarGetLeastF (const AstarGetLeastFParam &param, const AstarGetLeastFInput &input, AstarGetLeastFOutput &output)

  *Astar*算法 找到估最路径度最小的点
- void AstarGetNextRoad (const AstarGetNextRoadParam &param, const AstarGetNextRoadInput &input, AstarGetNextRoadOutput &output)

  *Astar*算法 找到下一条路

量

- AstarMapToAstarParam **param_map2astar** {m, as}
- AstarMapToAstarInput **input_map2astar** {}
- AstarMapToAstarOutput **output_map2astar** {as}
- AstarGetPathParam **param_get_path** {as}
- AstarGetPathInput **input_get_path** {origin, destination, originPointId, destinationPointId}
- AstarGetPathOutput **output_get_path** {path, as}
- AstarInitRoadParam **param_init_road** {as}
- AstarInitRoadInput **input_init_road** {number, xStart, yStart, xEnd, yEnd, length}
- AstarInitRoadOutput **output_init_road** {as}
- AstarInitLinkParam **param_init_link** {as}
- AstarInitLinkInput **input_init_link** {from, to}
- AstarInitLinkOutput **output_init_link** {as}
- AstarResetParam **param_reset** {as}
- AstarResetInput **input_reset** {}
- AstarResetOutput **output_reset** {as}
- AstarModuleSelfCheckParam **param_module_selfcheck** {as}
- AstarModuleSelfCheckInput **input_module_selfcheck** {path}
- AstarModuleSelfCheckOutput **output_module_selfcheck** {}
- AstarModuleSelfCheckPrintParam **param_module_selfcheck_print** {as}
- AstarModuleSelfCheckPrintInput **input_module_selfcheck_print** {pathLanes}
- AstarModuleSelfCheckPrintOutput **output_module_selfcheck_print** {}
- AstarFindLaneParam **param_find_lane** {as}
- AstarFindLaneInput **input_find_lane** {m, path}
- AstarFindLaneOutput **output_find_lane** {as}
- AstarFindLaneRParam **param_find_lane_r** {as}
- AstarFindLaneRInput **input_find_lane_r** {roads, path, pathLanes, lane_id}
- AstarFindLaneROutput **output_find_lane_r** {as}
- AstarRecursiveFindLaneParam **param_recursive_find_lane** {as}
- AstarRecursiveFindLaneInput **input_recursive_find_lane** {roads, path, optional_lanes, road_id, lane_id}
- AstarRecursiveFindLaneOutput **output_recursive_find_lane** {as}

4.2.1 AstarCalcF()

```
void GolbalPlanning::AstarCalcF (
            const AstarCalcFParam & param,
            const AstarCalcFInput & input,
            AstarCalcFOutput & output )
```

Astar算法 算当前点估最路径度

参数

[IN]	param Astar
[IN]	input
[OUT]	output F

@cn_name: Astar算法的主要部分

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.2.2 AstarCalcG()

```
void GolbalPlanning::AstarCalcG (
            const AstarCalcGParam & param,
            const AstarCalcGInput & input,
            AstarCalcGOutput & output )
```

Astar算法 算起点到当前点的已知度

参数

[IN]	param Astar
[IN]	input 出点 当前点
[OUT]	output G

@cn_name: Astar算法的主要部分

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.2.3 AstarCalcH()

```
void GolbalPlanning::AstarCalcH (
            const AstarCalcHParam & param,
            const AstarCalcHInput & input,
            AstarCalcHOutput & output )
```

Astar算法 算当前点到点的估度

参数

[IN]	param Astar
[IN]	input 当前点 点
[OUT]	output H

@cn_name: Astar算法的主要部分

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.2.4 AstarFindLane()

```
void GolbalPlanning::AstarFindLane (
            const AstarFindLaneParam & param,
            const AstarFindLaneInput & input,
            AstarFindLaneOutput & output )
```

根据Astar道路全局划果，得道全局划果

参数

[IN]	param Astar
[IN]	input 地、道路全局划果
[OUT]	output 更新的Astar

@cn_name: Astar得全局划道果

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.2.5 AstarFindLaneR()

```
void GolbalPlanning::AstarFindLaneR (
            const AstarFindLaneRParam & param,
            const AstarFindLaneRInput & input,
            AstarFindLaneROutput & output )
```

根据Astar道路全局划果，使用的方法得道全局划果

参数

[IN]	param Astar
[IN]	input 道路向量数、道路全局划果、存果的向量数、点道Id
[OUT]	output 更新的Astar

@cn_name: Astar地得全局划道果

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.2.6 AstarFindPath()

```
void GolbalPlanning::AstarFindPath (
            const AstarFindPathParam & param,
```

```
            const AstarFindPathInput & input,
            AstarFindPathOutput & output )
```

Astar算法的主要部分

参数

[IN]	param Astar
[IN]	input 出点 目的地
[OUT]	output 合理的道路

@cn_name: Astar算法的主要部分

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

4.2.7 AstarGetLeastF()

```
void GolbalPlanning::AstarGetLeastF (
            const AstarGetLeastFParam & param,
            const AstarGetLeastFInput & input,
            AstarGetLeastFOutput & output )
```

Astar算法 找到估最路径度最小的点

参数

[IN]	param Astar
[IN]	input
[OUT]	output openList下

@cn_name: Astar算法的主要部分

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

4.2.8 AstarGetNextRoad()

```
void GolbalPlanning::AstarGetNextRoad (
            const AstarGetNextRoadParam & param,
            const AstarGetNextRoadInput & input,
            AstarGetNextRoadOutput & output )
```

Astar算法 找到下一条路

参数

[IN]	param Astar
[IN]	input road id
[OUT]	output 下一条路id

@cn_name: Astar算法的主要部分

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.2.9 AstarGetPath()

```
void GolbalPlanning::AstarGetPath (
            const AstarGetPathParam & param,
            const AstarGetPathInput & input,
            AstarGetPathOutput & output )
```

使用Astar算法取路径，包括理、后理等部分

参数

[IN]	param Astar
[IN]	input 起点、点、起点Id、点Id
[OUT]	output 更新的Astar，起点到点的路径

@cn_name: Astar取全局路径

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.2.10 AstarInitLink()

```
void GolbalPlanning::AstarInitLink (
            const AstarInitLinkParam & param,
            const AstarInitLinkInput & input,
            AstarInitLinkOutput & output )
```

Astar道路接系初始化

参数

[IN]	param Astar
[IN]	input 起始道路Id，到达道路Id
[OUT]	output 更新的Astar

@cn_name: Astar添加道路接系

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.2.11 AstarInitRoad()

```
void GolbalPlanning::AstarInitRoad (
            const AstarInitRoadParam & param,
            const AstarInitRoadInput & input,
            AstarInitRoadOutput & output )
```

Astar道路初始化

参数

[IN]	param Astar
[IN]	input 道路Id，起点XY，点XY，度
[OUT]	output 更新的Astar

@cn_name: Astar添加道路

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.2.12 AstarMapToAstar()

```
void GolbalPlanning::AstarMapToAstar (
            const AstarMapToAstarParam & param,
            const AstarMapToAstarInput & input,
            AstarMapToAstarOutput & output )
```

使用Map构建Astar

参数

[IN]	param 地，初始化的Astar
[IN]	input 无
[OUT]	output 更新的Astar

@cn_name: 使用地构建Astar

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.2.13 AstarModuleSelfCheck()

```
void GolbalPlanning::AstarModuleSelfCheck (
            const AstarModuleSelfCheckParam & param,
            const AstarModuleSelfCheckInput & input,
            AstarModuleSelfCheckOutput & output )
```

Astar道路Id

参数

[IN]	param Astar
[IN]	input 无
[OUT]	output 无

@cn_name: Astar道路Id

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.2.14 AstarModuleSelfCheckPrint()

```
void GolbalPlanning::AstarModuleSelfCheckPrint (
            const AstarModuleSelfCheckPrintParam & param,
            const AstarModuleSelfCheckPrintInput & input,
            AstarModuleSelfCheckPrintOutput & output )
```

Astar道路Id打印

参数

[IN]	param Astar
[IN]	input 无
[OUT]	output 无

@cn_name: Astar道路Id打印

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.2.15 AstarRecursiveFindLane()

```
void GolbalPlanning::AstarRecursiveFindLane (
            const AstarRecursiveFindLaneParam & param,
            const AstarRecursiveFindLaneInput & input,
            AstarRecursiveFindLaneOutput & output )
```

Astar得道全局划果主体函数

参数

[IN]	param Astar
[IN]	input 道路、道路全局划果、可道、点道路Id、点道Id
[OUT]	output 更新的Astar

@cn_name: Astar得道全局划果主体函数

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.2.16 AstarReset()

```
void GolbalPlanning::AstarReset (
            const AstarResetParam & param,
            const AstarResetInput & input,
            AstarResetOutput & output )
```

在保留地的情况下，初始化Astar参数

参数

[IN]	param Astar
[IN]	input 无
[OUT]	output 更新的Astar

@cn_name: Astar参数初始化

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.3 interp_m 功能模

是一个名interp_m的namespace。

- struct interpConstAccInput
- struct interpConstAccOutput
- struct interpConstAccParam
- struct interpLinearInput
- struct interpLinearOutput
- struct interpLinearParam

函数

2 个子模

- void linear (const interpLinearParam &param, const interpLinearInput &input, interpLinearOutput &output)

  根据初、和数，行性插
- void constant_acceleration (const interpConstAccParam &param, const interpConstAccInput &input, interpConstAccOutput &output)

  在等隔的情况下，行匀加速度插

4.3.1 constant_acceleration()

```
void interp_m::constant_acceleration (
            const interpConstAccParam & param,
            const interpConstAccInput & input,
            interpConstAccOutput & output )
```

在等隔的情况下，行匀加速度插

参数

[IN]	param 无
[IN]	input 初、、数、加速度
[OUT]	output 存插的向量数

@cn_name: 匀加速度插

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

4.3.2 linear()

```
void interp_m::linear (
            const interpLinearParam & param,
```

```
const interpLinearInput & input,
interpLinearOutput & output )
```

根据初、和数，行性插

参数

[IN]	param 无
[IN]	input 初、、数
[OUT]	output 存插的向量数

@cn_name: 性插

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

4.4 LocalPlanning 命名空参考

是一个名LocalPlanning的namespace。

- struct ableToPassYellowLightInput
- struct ableToPassYellowLightOutput
- struct ableToPassYellowLightParam
- struct angleRegulateInput
- struct angleRegulateOutput
- struct angleRegulateParam
- struct assessTrajectoryInput
- struct assessTrajectoryOutput
- struct assessTrajectoryParam
- struct boundaryCollisionCheckInput
- struct boundaryCollisionCheckOutput
- struct boundaryCollisionCheckParam
- struct calculateCurvatureInput
- struct calculateCurvatureOutput
- struct calculateCurvatureParam
- struct changeCurrentPointInput
- struct changeCurrentPointOutput
- struct changeCurrentPointParam
- struct changeLocalToGlobalInput
- struct changeLocalToGlobalOutput
- struct changeLocalToGlobalParam
- struct clearPathListInput
- struct clearPathListOutput
- struct clearPathListParam
- struct coordTran2DInput

- struct coordTran2DOutput
- struct coordTran2DParam
- struct courseAngleReviseInput
- struct courseAngleReviseOutput
- struct courseAngleReviseParam
- struct cruiseControllerInput
- struct cruiseControllerOutput
- struct cruiseControllerParam
- struct DecisionData
- struct findMaxInput
- struct findMaxOutput
- struct findMaxParam
- struct findMinInput
- struct findMinOutput
- struct findMinParam
- struct fixRoadLaneIndexInput
- struct fixRoadLaneIndexOutput
- struct fixRoadLaneIndexParam
- struct frenet2CartesianInput
- struct frenet2CartesianOutput
- struct frenet2CartesianParam
- struct gaussConvertInput
- struct gaussConvertOutput
- struct gaussConvertParam
- struct generateBezierPathInFrenetInput
- struct generateBezierPathInFrenetOutput
- struct generateBezierPathInFrenetParam
- struct generateBezierPathListInFrenetInput
- struct generateBezierPathListInFrenetOutput
- struct generateBezierPathListInFrenetParam
- struct generateSegmentedBezierPathListInFrenetInput
- struct generateSegmentedBezierPathListInFrenetOutput
- struct generateSegmentedBezierPathListInFrenetParam
- struct generateSpeedListInput
- struct generateSpeedListOutput
- struct generateSpeedListParam
- struct generateTrajectoryListInput
- struct generateTrajectoryListOutput
- struct generateTrajectoryListParam
- struct getAngleDiffInput
- struct getAngleDiffOutput
- struct getAngleDiffParam
- struct getAngleInput
- struct getAngleOutput
- struct getAngleParam
- struct getCurrentPositionInput
- struct getCurrentPositionOutput
- struct getCurrentPositionParam
- struct getDistanceInput
- struct getDistanceOutput
- struct getDistanceParam
- struct getFrenetLocationPointInput
- struct getFrenetLocationPointOutput
- struct getFrenetLocationPointParam
- struct getMinDistanceOfPointInput

- struct getMinDistanceOfPointOutput
- struct getMinDistanceOfPointParam
- struct getNextRoadLaneIdInput
- struct getNextRoadLaneIdOutput
- struct getNextRoadLaneIdParam
- struct getOptimalTrajectoryIndexInput
- struct getOptimalTrajectoryIndexOutput
- struct getOptimalTrajectoryIndexParam
- struct getPlanningDistanceInput
- struct getPlanningDistanceOutput
- struct getPlanningDistanceParam
- struct getPlanningPointInput
- struct getPlanningPointOutput
- struct getPlanningPointParam
- struct getReferenceLineInput
- struct getReferenceLineOutput
- struct getReferenceLineParam
- struct getTrajectoryPlanningResultInput
- struct getTrajectoryPlanningResultOutput
- struct getTrajectoryPlanningResultParam
- struct hasPrecedingVehicleInput
- struct hasPrecedingVehicleOutput
- struct hasPrecedingVehicleParam
- struct inAreaInput
- struct inAreaOutput
- struct inAreaParam
- struct initLocalPlanningInput
- struct initLocalPlanningOutput
- struct initLocalPlanningParam
- struct innerCollisionCheckInput
- struct innerCollisionCheckOutput
- struct innerCollisionCheckParam
- struct limitPlanningDistanceInput
- struct limitPlanningDistanceOutput
- struct limitPlanningDistanceParam
- struct loadStopPointsFromYamlInput
- struct loadStopPointsFromYamlOutput
- struct loadStopPointsFromYamlParam
- struct loadStopPointsInput
- struct loadStopPointsOutput
- struct loadStopPointsParam
- struct PlanningPoint
- struct PlanningTrajectory
- struct pointCollisionCheckInput
- struct pointCollisionCheckOutput
- struct pointCollisionCheckParam
- struct pointDistanceInput
- struct pointDistanceOutput
- struct pointDistanceParam
- struct pointOnCubicBezierInput
- struct pointOnCubicBezierOutput
- struct pointOnCubicBezierParam
- struct pointToLineDistanceInput
- struct pointToLineDistanceOutput
- struct pointToLineDistanceParam

- struct processTrafficLightInput
- struct processTrafficLightOutput
- struct processTrafficLightParam
- struct ReferenceLine
- struct restartInput
- struct restartOutput
- struct restartParam
- struct stopInput
- struct stopOutput
- struct stopParam
- struct stopPointJudgeInput
- struct stopPointJudgeOutput
- struct stopPointJudgeParam
- struct trajectoryCollisionCheckInput
- struct trajectoryCollisionCheckOutput
- struct trajectoryCollisionCheckParam

函数

48 个子模

- void gaussConvert (const gaussConvertParam &param, const gaussConvertInput &input, gaussConvertOutput &output)

  度全局坐
- void coordTran2D (const coordTran2DParam &param, const coordTran2DInput &input, coordTran2DOutput &output)

  全局坐局部坐
- void inArea (const inAreaParam &param, const inAreaInput &input, inAreaOutput &output)

  判断点是否在区域内
- void pointDistance (const pointDistanceParam &param, const pointDistanceInput &input, pointDistanceOutput &output)

  算两点之的距离
- void courseAngleRevise (const courseAngleReviseParam &param, const courseAngleReviseInput &input, courseAngleReviseOutput &output)

  逆旋坐系*90*度，向*0*度成向北*0*度
- void loadStopPoints (const loadStopPointsParam &param, const loadStopPointsInput &input, loadStopPointsOutput &output)

  从文件中加停止点
- void loadStopPointsFromYaml (const loadStopPointsFromYamlParam &param, const loadStopPointsFromYamlInput &input, loadStopPointsFromYamlOutput &output)

  从文件中加停止点
- void getNextRoadLaneId (const getNextRoadLaneIdParam &param, const getNextRoadLaneIdInput &input, getNextRoadLaneIdOutput &output)

  从全局划中，取后道和道路*Id*
- void getPlanningDistance (const getPlanningDistanceParam &param, const getPlanningDistanceInput &input, getPlanningDistanceOutput &output)

  根据速度、曲率取划距离
- void getPlanningPoint (const getPlanningPointParam &param, const getPlanningPointInput &input, getPlanningPointOutput &output)

  根据信息取划瞄点
- void limitPlanningDistance (const limitPlanningDistanceParam &param, const limitPlanningDistanceInput &input, limitPlanningDistanceOutput &output)

限制刹距离，防止突

- void getAngle (const getAngleParam &param, const getAngleInput &input, getAngleOutput &output)

  算坐系中两点角度

- void getAngleDiff (const getAngleDiffParam &param, const getAngleDiffInput &input, getAngleDiffOutput &output)

  算两角度之差，并正化

- void **localPlanning** (const pc::Imu &imu, double velocity, const RoadMap &map, DecisionData &decision↩ Data, const prediction::ObjectList &predictionMsg, const std::vector< std::tuple< int32_t, int32_t >> &routingList, const std::vector< GaussRoadPoint > stopPoints, const infopack::TrafficLight &trafficLight, std::vector< infopack::ObjectsProto > objectsCmd)
- void restart (const restartParam &param, const restartInput &input, restartOutput &output)

  停后，判断能否重新后

- void stop (const stopParam &param, const stopInput &input, stopOutput &output)

  停修改速度曲

- void stopPointJudge (const stopPointJudgeParam &param, const stopPointJudgeInput &input, stopPointJudgeOutput &output)

  判断是否在停区域

- void pointOnCubicBezier (const pointOnCubicBezierParam &param, const pointOnCubicBezierInput &input, pointOnCubicBezierOutput &output)

  算塞曲上的点

- void generateBezierPathInFrenet (const generateBezierPathInFrenetParam &param, const generateBezierPathInFrenetInput &input, generateBezierPathInFrenetOutput &output)

  在*frenet*中生成塞曲

- void clearPathList (const clearPathListParam &param, const clearPathListInput &input, clearPathListOutput &output)

  清除上次生成的迹

- void getTrajectoryPlanningResult (const getTrajectoryPlanningResultParam &param, const getTrajectoryPlanningResultInput &input, getTrajectoryPlanningResultOutput &output)

  完成了多条迹和速度曲的生成，并化到笛卡坐系中，作出了迹的

- void getDistance (const getDistanceParam &param, const getDistanceInput &input, getDistanceOutput &output)

  算两点之的距离

- void pointToLineDistance (const pointToLineDistanceParam &param, const pointToLineDistanceInput &input, pointToLineDistanceOutput &output)

  算点到段的距离

- void angleRegulate (const angleRegulateParam &param, const angleRegulateInput &input, angleRegulateOutput &output)

  化角度到*0-180*中，便于比

- void getCurrentPosition (const getCurrentPositionParam &param, const getCurrentPositionInput &input, getCurrentPositionOutput &output)

  找距离位置最近的路点

- void generateSpeedList (const generateSpeedListParam &param, const generateSpeedListInput &input, generateSpeedListOutput &output)

  根据道路信息等，采生成速度曲

- void generateTrajectoryList (const generateTrajectoryListParam &param, const generateTrajectoryListInput &input, generateTrajectoryListOutput &output)

  生成采迹，供后

- void processTrafficLight (const processTrafficLightParam &param, const processTrafficLightInput &input, processTrafficLightOutput &output)

  根据灯信息更新决策数据

- void getOptimalTrajectoryIndex (const getOptimalTrajectoryIndexParam &param, const getOptimalTrajectoryIndexInput &input, getOptimalTrajectoryIndexOutput &output)

  估迹，并取最迹索引

- void trajectoryCollisionCheck (const trajectoryCollisionCheckParam &param, const trajectoryCollisionCheckInput &input, trajectoryCollisionCheckOutput &output)

  迹碰撞，*true*表示可行，*false*表示碰撞
- void ableToPassYellowLight (const ableToPassYellowLightParam &param, const ableToPassYellowLightInput &input, ableToPassYellowLightOutput &output)

  判断是否能通黄灯
- void pointCollisionCheck (const pointCollisionCheckParam &param, const pointCollisionCheckInput &input, pointCollisionCheckOutput &output)

  迹点碰撞
- void boundaryCollisionCheck (const boundaryCollisionCheckParam &param, const boundaryCollisionCheckInput &input, boundaryCollisionCheckOutput &output)

  碰撞
- void innerCollisionCheck (const innerCollisionCheckParam &param, const innerCollisionCheckInput &input, innerCollisionCheckOutput &output)

  内部包含
- void changeCurrentPoint (const changeCurrentPointParam &param, const changeCurrentPointInput &input, changeCurrentPointOutput &output)

  判断是否可以道
- void changeLocalToGlobal (const changeLocalToGlobalParam &param, const changeLocalToGlobalInput &input, changeLocalToGlobalOutput &output)

  局部坐迹全局迹
- void initLocalPlanning (const initLocalPlanningParam &param, const initLocalPlanningInput &input, initLocalPlanningOutput &output)

  局部划决策数据初始化
- void fixRoadLaneIndex (const fixRoadLaneIndexParam &param, const fixRoadLaneIndexInput &input, fixRoadLaneIndexOutput &output)

  返回*decisionDate*中*roadID* 和*laneID*的索引
- void calculateCurvature (const calculateCurvatureParam &param, const calculateCurvatureInput &input, calculateCurvatureOutput &output)

  由相三点近似算曲率
- void assessTrajectory (const assessTrajectoryParam &param, const assessTrajectoryInput &input, assessTrajectoryOutput &output)

  返回距离障碍物最近距离，行迹估
- void getMinDistanceOfPoint (const getMinDistanceOfPointParam &param, const getMinDistanceOfPointInput &input, getMinDistanceOfPointOutput &output)

  返回点距离障碍物最近距离
- void findMin (const findMinParam &param, const findMinInput &input, findMinOutput &output)

  返回*double*数最小
- void findMax (const findMaxParam &param, const findMaxInput &input, findMaxOutput &output)

  返回*double*数最大
- void getReferenceLine (const getReferenceLineParam &param, const getReferenceLineInput &input, getReferenceLineOutput &output)

  取局部划参考迹
- void getFrenetLocationPoint (const getFrenetLocationPointParam &param, const getFrenetLocationPointInput &input, getFrenetLocationPointOutput &output)

  取当前位置在*frenet*坐系中坐
- void frenet2Cartesian (const frenet2CartesianParam &param, const frenet2CartesianInput &input, frenet2CartesianOutput &output)

  将*frenet*坐系坐自坐系坐，*xy*右前
- void generateBezierPathListInFrenet (const generateBezierPathListInFrenetParam &param, const generateBezierPathListInFrenetInput &input, generateBezierPathListInFrenetOutput &output)

  在*frenet*坐系中生成多条塞曲

- void hasPrecedingVehicle (const hasPrecedingVehicleParam &param, const hasPrecedingVehicleInput &input, hasPrecedingVehicleOutput &output)

  判断前方是否有需要跟随的
- void cruiseController (const cruiseControllerParam &param, const cruiseControllerInput &input, cruiseControllerOutput &output)

  *acc*景下，目速度算
- void **generateSegmentedBezierPathListInFrenet** (const generateSegmentedBezierPathListInFrenetParam &param, const generateSegmentedBezierPathListInFrenetInput &input, generateSegmentedBezierPathListInFrenetOutput &output)

4.4.1 ableToPassYellowLight()

```
void LocalPlanning::ableToPassYellowLight (
            const ableToPassYellowLightParam & param,
            const ableToPassYellowLightInput & input,
            ableToPassYellowLightOutput & output )
```

判断是否能通黄灯

参数

[IN]	param 无
[IN]	input 迹，剩余，到停止前的距离
[OUT]	output 是否通

@cn_name: 判断黄灯能否通

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.2 angleRegulate()

```
void LocalPlanning::angleRegulate (
            const angleRegulateParam & param,
            const angleRegulateInput & input,
            angleRegulateOutput & output )
```

化角度到0-180中，便于比

参数

[IN]	param 无
[IN]	input 原角度
[OUT]	output 化后的角度

@cn_name: 正化角度

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.3 assessTrajectory()

```
void LocalPlanning::assessTrajectory (
            const assessTrajectoryParam & param,
            const assessTrajectoryInput & input,
            assessTrajectoryOutput & output )
```

返回距离障碍物最近距离，行迹估

参数

[IN]	param 格障碍物，imu，路障碍物
[IN]	input 迹
[OUT]	output 距离障碍物最近距离

@cn_name: 迹估

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.4 boundaryCollisionCheck()

```
void LocalPlanning::boundaryCollisionCheck (
            const boundaryCollisionCheckParam & param,
            const boundaryCollisionCheckInput & input,
            boundaryCollisionCheckOutput & output )
```

碰撞

参数

[IN]	param 无
[IN]	input 段点
[OUT]	output 是否碰撞

@cn_name: 碰撞

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.5 calculateCurvature()

```
void LocalPlanning::calculateCurvature (
            const calculateCurvatureParam & param,
            const calculateCurvatureInput & input,
            calculateCurvatureOutput & output )
```

由相三点近似算曲率

参数

[IN]	param 无
[IN]	input 点，前一点，后一点
[OUT]	output 曲率

@cn_name: 算曲率

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.6 changeCurrentPoint()

```
void LocalPlanning::changeCurrentPoint (
            const changeCurrentPointParam & param,
            const changeCurrentPointInput & input,
            changeCurrentPointOutput & output )
```

判断是否可以道

参数

[IN]	param 当前道路，当前道
[IN]	input 原划瞄点道Id，原划瞄点路点Id，原划瞄点，向左是向右道
[OUT]	output 新瞄点道Id，新划瞄点路点Id，新划瞄点，是否道

@cn_name: 判断是否可以道

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.7 changeLocalToGlobal()

```
void LocalPlanning::changeLocalToGlobal (
            const changeLocalToGlobalParam & param,
            const changeLocalToGlobalInput & input,
            changeLocalToGlobalOutput & output )
```

局部坐迹全局迹

参数

[IN]	param 无
[IN]	input 局部坐迹，局部坐迹原地的全局点
[OUT]	output 全局迹

@cn_name: 局部坐迹全局坐迹

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.8 clearPathList()

```
void LocalPlanning::clearPathList (
            const clearPathListParam & param,
            const clearPathListInput & input,
            clearPathListOutput & output )
```

清除上次生成的迹

参数

[IN]	param 无
[IN]	input 存迹的向量
[OUT]	output 判断果

@cn_name: 清除迹

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.9 coordTran2D()

```
void LocalPlanning::coordTran2D (
            const coordTran2DParam & param,
```

```
            const coordTran2DInput & input,
            coordTran2DOutput & output )
```

全局坐局部坐

参数

[IN]	param 无
[IN]	input 全局坐度、角度，初始化局部坐xy、角度
[OUT]	output 局部坐、角度

@cn_name: 全局坐自局部坐

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

4.4.10 courseAngleRevise()

```
void LocalPlanning::courseAngleRevise (
            const courseAngleReviseParam & param,
            const courseAngleReviseInput & input,
            courseAngleReviseOutput & output )
```

逆旋坐系90度，向0度成向北0度

参数

[IN]	param 无
[IN]	input 原始角度
[OUT]	output 化角度

@cn_name: 坐系逆旋90度

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

4.4.11 cruiseController()

```
void LocalPlanning::cruiseController (
            const cruiseControllerParam & param,
            const cruiseControllerInput & input,
            cruiseControllerOutput & output )
```

acc景下，目速度算

参数

[IN]	param imu，路障碍物
[IN]	input 前索引，距前距离
[OUT]	output 期速度

@cn_name: 算acc目速度

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.12 findMax()

```
void LocalPlanning::findMax (
            const findMaxParam & param,
            const findMaxInput & input,
            findMaxOutput & output )
```

返回double数最大

参数

[IN]	param 无
[IN]	input double数
[OUT]	output double数最大

@cn_name: 返回double数最大

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.13 findMin()

```
void LocalPlanning::findMin (
            const findMinParam & param,
            const findMinInput & input,
            findMinOutput & output )
```

返回double数最小

参数

[IN]	param 无
[IN]	input double数
[OUT]	output double数最小

@cn_name: 返回double数最小

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

4.4.14 fixRoadLaneIndex()

```
void LocalPlanning::fixRoadLaneIndex (
            const fixRoadLaneIndexParam & param,
            const fixRoadLaneIndexInput & input,
            fixRoadLaneIndexOutput & output )
```

返回decisionDate中roadID 和laneID的索引

参数

[IN]	param 地
[IN]	input 决策数据
[OUT]	output roadID 和laneID的索引

@cn_name: 得道路和道的索引

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

4.4.15 frenet2Cartesian()

```
void LocalPlanning::frenet2Cartesian (
            const frenet2CartesianParam & param,
            const frenet2CartesianInput & input,
            frenet2CartesianOutput & output )
```

将frenet坐系坐自坐系坐，xy右前

参数

[IN]	param 无
[IN]	input 坐点sl，初始化xy，参考迹，上次搜索的索引，本次需返回的坐点的索引
[OUT]	output 坐点xy，坐点的索引

@cn_name: frenet坐系局部坐系

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

4.4.16 gaussConvert()

```
void LocalPlanning::gaussConvert (
            const gaussConvertParam & param,
            const gaussConvertInput & input,
            gaussConvertOutput & output )
```

度全局坐

参数

[IN]	param :无
[IN]	input :坐度，初始化全局坐xy
[OUT]	output :全局坐

@cn_name: 度全局坐

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

4.4.17 generateBezierPathInFrenet()

```
void LocalPlanning::generateBezierPathInFrenet (
            const generateBezierPathInFrenetParam & param,
            const generateBezierPathInFrenetInput & input,
            generateBezierPathInFrenetOutput & output )
```

在frenet中生成塞曲

参数

[IN]	param 无
[IN]	input 起点、点、初始化迹
[OUT]	output 生成迹

@cn_name: 生成塞曲

@granularity: composite //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

4.4.18 generateBezierPathListInFrenet()

```
void LocalPlanning::generateBezierPathListInFrenet (
            const generateBezierPathListInFrenetParam & param,
            const generateBezierPathListInFrenetInput & input,
            generateBezierPathListInFrenetOutput & output )
```

在frenet坐系中生成多条塞曲

参数

[IN]	param 无
[IN]	input 参考迹，在frenet坐系下坐，初始化的迹向量数
[OUT]	output 迹向量数

@cn_name: 在frenet坐系中生成多条塞曲

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: plannings

4.4.19 generateSpeedList()

```
void LocalPlanning::generateSpeedList (
            const generateSpeedListParam & param,
            const generateSpeedListInput & input,
            generateSpeedListOutput & output )
```

根据道路信息等，采生成速度曲

参数

[IN]	param 路障碍物，imu，停止点
[IN]	input 速，决策数据
[OUT]	output 更新后的决策数据

@cn_name: 生成速度曲

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.20 generateTrajectoryList()

```
void LocalPlanning::generateTrajectoryList (
            const generateTrajectoryListParam & param,
```

```
            const generateTrajectoryListInput & input,
            generateTrajectoryListOutput & output )
```

生成采迹，供后

参数

[IN]	param 无
[IN]	input 决策数据
[OUT]	output 更新后的决策数据

@cn_name: 生成采曲

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

4.4.21 getAngle()

```
void LocalPlanning::getAngle (
            const getAngleParam & param,
            const getAngleInput & input,
            getAngleOutput & output )
```

算坐系中两点角度

参数

[IN]	param 无
[IN]	input 两点XY坐
[OUT]	output 两点角度

@cn_name: 算两点角度

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

4.4.22 getAngleDiff()

```
void LocalPlanning::getAngleDiff (
            const getAngleDiffParam & param,
            const getAngleDiffInput & input,
            getAngleDiffOutput & output )
```

算两角度之差，并正化

参数

[IN]	param 无
[IN]	input 两角度
[O↩ UT]	output 两角度之差

@cn_name: 算两角度之差

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.23 getCurrentPosition()

```
void LocalPlanning::getCurrentPosition (
            const getCurrentPositionParam & param,
            const getCurrentPositionInput & input,
            getCurrentPositionOutput & output )
```

找距离位置最近的路点

参数

[IN]	param imu，地
[IN]	input 决策数据
[O↩ UT]	output 更新的决策具

@cn_name: 找距离位置最近的路点

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.24 getDistance()

```
void LocalPlanning::getDistance (
            const getDistanceParam & param,
            const getDistanceInput & input,
            getDistanceOutput & output )
```

算两点之的距离

参数

[IN]	param 无
[IN]	input 两点XY坐
[O↩ UT]	output 两点之的距离

@cn_name: 算两点之的距离

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

4.4.25 getFrenetLocationPoint()

```
void LocalPlanning::getFrenetLocationPoint (
            const getFrenetLocationPointParam & param,
            const getFrenetLocationPointInput & input,
            getFrenetLocationPointOutput & output )
```

取当前位置在frenet坐系中坐

参数

[IN]	param 无
[IN]	input 决策数据，参考迹，当前位置
[OUT]	output 更新后的决策数据

@cn_name: 取在frenet坐系中坐

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

4.4.26 getMinDistanceOfPoint()

```
void LocalPlanning::getMinDistanceOfPoint (
            const getMinDistanceOfPointParam & param,
            const getMinDistanceOfPointInput & input,
            getMinDistanceOfPointOutput & output )
```

返回点距离障碍物最近距离

参数

[IN]	param 格障碍物，imu，路障碍物
[IN]	input 点
[OUT]	output 距离障碍物最近距离

@cn_name: 算点距离障碍物最近距离（一般景）

@granularity: composite //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

4.4.27 getNextRoadLaneId()

```
void LocalPlanning::getNextRoadLaneId (
            const getNextRoadLaneIdParam & param,
            const getNextRoadLaneIdInput & input,
            getNextRoadLaneIdOutput & output )
```

从全局划中，取后道和道路Id

参数

[IN]	param 存全局划道和道路的向量
[IN]	input 自当前道和道路Id
[OUT]	output 后道和道路Id

@cn_name: 取后道和道路Id

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.28 getOptimalTrajectoryIndex()

```
void LocalPlanning::getOptimalTrajectoryIndex (
            const getOptimalTrajectoryIndexParam & param,
            const getOptimalTrajectoryIndexInput & input,
            getOptimalTrajectoryIndexOutput & output )
```

估迹，并取最迹索引

参数

[IN]	param 格障碍物，交通灯，imu，路障碍物
[IN]	input 决策数据
[OUT]	output 更新后的决策数据，最迹索引

@cn_name: 取最迹索引

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.29 getPlanningDistance()

```
void LocalPlanning::getPlanningDistance (
            const getPlanningDistanceParam & param,
            const getPlanningDistanceInput & input,
            getPlanningDistanceOutput & output )
```

根据速度、曲率取划距离

参数

[IN]	param 无
[IN]	input 速度、曲率
[OUT]	output 划距离

@cn_name: 算划距离

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.30 getPlanningPoint()

```
void LocalPlanning::getPlanningPoint (
            const getPlanningPointParam & param,
            const getPlanningPointInput & input,
            getPlanningPointOutput & output )
```

根据信息取划瞄点

参数

[IN]	param 当前位置，决策数据，地
[IN]	input 划距离，朝向角，后道Id
[OUT]	output 划瞄点

@cn_name: 算划瞄点

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.31 getReferenceLine()

```
void LocalPlanning::getReferenceLine (
            const getReferenceLineParam & param,
```

```
            const getReferenceLineInput & input,
            getReferenceLineOutput & output )
```

取局部划参考迹

参数

[IN]	param 地
[IN]	input 决策数据，当前位置
[OUT]	output 参考迹

@cn_name: 取参考迹

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.32 getTrajectoryPlanningResult()

```
void LocalPlanning::getTrajectoryPlanningResult (
            const getTrajectoryPlanningResultParam & param,
            const getTrajectoryPlanningResultInput & input,
            getTrajectoryPlanningResultOutput & output )
```

完成了多条迹和速度曲的生成，并化到笛卡坐系中，作出了迹的

参数

[IN]	param imu，格障碍物，地，交通灯，路障碍物
[IN]	input 决策数据
[OUT]	output 更新后的决策数据，是否成功生成迹

@cn_name: 取迹划果

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.33 hasPrecedingVehicle()

```
void LocalPlanning::hasPrecedingVehicle (
            const hasPrecedingVehicleParam & param,
            const hasPrecedingVehicleInput & input,
            hasPrecedingVehicleOutput & output )
```

判断前方是否有需要跟随的

参数

[IN]	param imu，路障碍物
[IN]	input 无
[OUT]	output 判断果

@cn_name: 判断是否有需要跟随的

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.34 inArea()

```
void LocalPlanning::inArea (
            const inAreaParam & param,
            const inAreaInput & input,
            inAreaOutput & output )
```

判断点是否在区域内

参数

[IN]	param 无
[IN]	input 目区域中心XY坐，判断点XY坐
[OUT]	output 判断果

@cn_name: 判断点是否在区域内

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.35 initLocalPlanning()

```
void LocalPlanning::initLocalPlanning (
            const initLocalPlanningParam & param,
            const initLocalPlanningInput & input,
            initLocalPlanningOutput & output )
```

局部划决策数据初始化

参数

[IN]	param 无
[IN]	input 决策数据
[OUT]	output 初始化后的决策数据

@cn_name: 初始化局部划

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

4.4.36 innerCollisionCheck()

```
void LocalPlanning::innerCollisionCheck (
            const innerCollisionCheckParam & param,
            const innerCollisionCheckInput & input,
            innerCollisionCheckOutput & output )
```

内部包含

参数

[IN]	param 无
[IN]	input 点和障碍物点
[OUT]	output 是否包含

@cn_name: 内部包含

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

4.4.37 limitPlanningDistance()

```
void LocalPlanning::limitPlanningDistance (
            const limitPlanningDistanceParam & param,
            const limitPlanningDistanceInput & input,
            limitPlanningDistanceOutput & output )
```

限制划距离，防止突

参数

[IN]	param 无
[IN]	input 划划距离，上次划距离
[OUT]	output 更新后的划距离

@cn_name: 限制划距离

@granularity: atomic //函数件的粒度， atomic： 基件（原子），composite>： 合件（在前端可以向下一展）

@tag: planning

4.4.38 loadStopPoints()

```
void LocalPlanning::loadStopPoints (
            const loadStopPointsParam & param,
            const loadStopPointsInput & input,
            loadStopPointsOutput & output )
```

从文件中加停止点

参数

[IN]	param 文件地址
[IN]	input 停止点、停止点路点、X偏移、Y偏移
[OUT]	output 更新后的停止点、更新后的停止点路点

@cn_name: 加停止点

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.39 loadStopPointsFromYaml()

```
void LocalPlanning::loadStopPointsFromYaml (
            const loadStopPointsFromYamlParam & param,
            const loadStopPointsFromYamlInput & input,
            loadStopPointsFromYamlOutput & output )
```

从文件中加停止点

参数

[IN]	param 地、文件地址
[IN]	input 停止点、停止点路点、X偏移、Y偏移
[OUT]	output 更新的停止点、更新的停止点路点

@cn_name: 从Yaml文件中加停止点

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.40 pointCollisionCheck()

```
void LocalPlanning::pointCollisionCheck (
            const pointCollisionCheckParam & param,
            const pointCollisionCheckInput & input,
            pointCollisionCheckOutput & output )
```

迹点碰撞

参数

[IN]	param 无
[IN]	input 点，障碍物中心点，障碍物
[OUT]	output 是否碰撞

@cn_name: 迹点碰撞

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.41 pointDistance()

```
void LocalPlanning::pointDistance (
            const pointDistanceParam & param,
            const pointDistanceInput & input,
            pointDistanceOutput & output )
```

算两点之的距离

参数

[IN]	param 无
[IN]	input 两点的XY坐
[OUT]	output 距离

@cn_name: 算两点之的距离

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.42 pointOnCubicBezier()

```
void LocalPlanning::pointOnCubicBezier (
            const pointOnCubicBezierParam & param,
```

```
            const pointOnCubicBezierInput & input,
            pointOnCubicBezierOutput & output )
```

算塞曲上的点

参数

[IN]	param 无
[IN]	input 塞曲控制点，参数
[OUT]	output 判断果

@cn_name: 算塞曲上的点

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.43 pointToLineDistance()

```
void LocalPlanning::pointToLineDistance (
            const pointToLineDistanceParam & param,
            const pointToLineDistanceInput & input,
            pointToLineDistanceOutput & output )
```

算点到段的距离

参数

[IN]	param 无
[IN]	input 段起点，段点，点
[OUT]	output 点到段的距离

@cn_name: 算点到段的距离

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.44 processTrafficLight()

```
void LocalPlanning::processTrafficLight (
            const processTrafficLightParam & param,
            const processTrafficLightInput & input,
            processTrafficLightOutput & output )
```

根据灯信息更新决策数据

参数

[IN]	param 格障碍物，交通灯
[IN]	input 决策数据
[OUT]	output 更新后的决策数据，灯信息

@cn_name: 理交通信号灯

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.45 restart()

```
void LocalPlanning::restart (
            const restartParam & param,
            const restartInput & input,
            restartOutput & output )
```

停后，判断能否重新后

参数

[IN]	param 格障碍物
[IN]	input 速
[OUT]	output 判断果

@cn_name: 判断能否重新后

@granularity: atomic //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.46 stop()

```
void LocalPlanning::stop (
            const stopParam & param,
            const stopInput & input,
            stopOutput & output )
```

停修改速度曲

参数

[IN]	param 无
[IN]	input 决策数据，当前速
[OUT]	output 更新的决策数据

@cn_name: 理停速度

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.47 stopPointJudge()

```
void LocalPlanning::stopPointJudge (
            const stopPointJudgeParam & param,
            const stopPointJudgeInput & input,
            stopPointJudgeOutput & output )
```

判断是否在停区域

参数

[IN]	param imu，停止点
[IN]	input 无
[OUT]	output 判断果

@cn_name: 判断停

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

@tag: planning

4.4.48 trajectoryCollisionCheck()

```
void LocalPlanning::trajectoryCollisionCheck (
            const trajectoryCollisionCheckParam & param,
            const trajectoryCollisionCheckInput & input,
            trajectoryCollisionCheckOutput & output )
```

迹碰撞，true表示可行，false表示碰撞

参数

[IN]	param 障碍物
[IN]	input 迹
[OUT]	output 是否碰撞

@cn_name: 迹碰撞

@granularity: composite //函数件的粒度， atomic：基件（原子），composite>：合件（在前端可以向下一展）

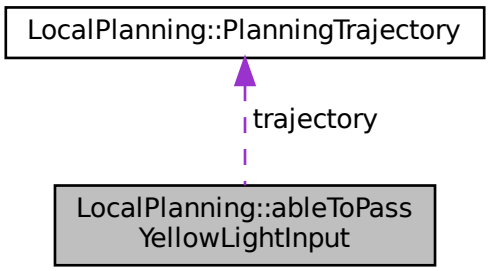
@tag: planning

Chapter 5

明

5.1 LocalPlanning::ableToPassYellowLightInput构体 参考

LocalPlanning::ableToPassYellowLightInput 的作:



Public 属性

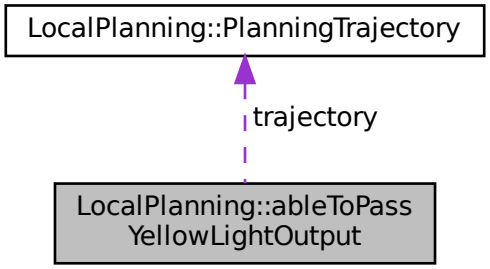
- PlanningTrajectory **trajectory**
- double **remaining_time**
- double **lane_length_before_intersection**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.2 LocalPlanning::ableToPassYellowLightOutput构体 参考

LocalPlanning::ableToPassYellowLightOutput 的作:



Public 属性

- PlanningTrajectory **trajectory**
- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.3 LocalPlanning::ableToPassYellowLightParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.4 LocalPlanning::angleRegulateInput构体 参考

Public 属性

- double **angleInput**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.5 LocalPlanning::angleRegulateOutput构体 参考

Public 属性

- double **angleOutput**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

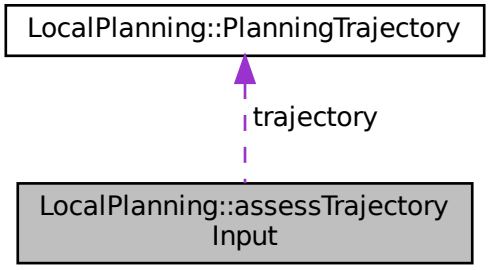
5.6 LocalPlanning::angleRegulateParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.7 LocalPlanning::assessTrajectoryInput构体 参考

LocalPlanning::assessTrajectoryInput 的作:



Public 属性

- PlanningTrajectory **trajectory**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.8 LocalPlanning::assessTrajectoryOutput构体 参考

Public 属性

- double **minDistance**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.9 LocalPlanning::assessTrajectoryParam构体 参考

Public 属性

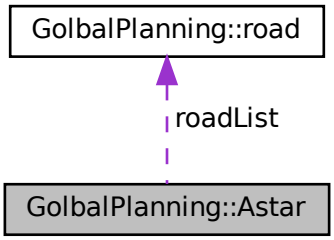
- prediction::ObjectList **prediction**
- pc::Imu **imu**
- std::vector< infopack::ObjectsProto > **objectsCmd**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.10 GolbalPlanning::Astar 参考

GolbalPlanning::Astar 的作:



Public 属性

- road * **roadList**
- list< int > **path**
- vector< pair< int, int > > **pathLanes**
- int **size** = 10000
- std::priority_queue< node, std::vector< node >, cmp > **openList**

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/src/global↩ Planning_m _2.cpp

5.11 GolbalPlanning::AstarCalcFInput构体 参考

Public 属性

- int **tmp**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.12 GolbalPlanning::AstarCalcFOutput构体 参考

Public 属性

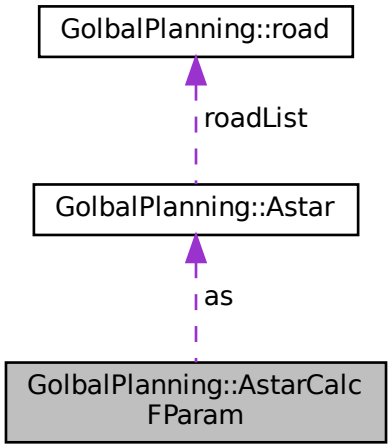
- double **result**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.13 GolbalPlanning::AstarCalcFParam构体 参考

GolbalPlanning::AstarCalcFParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.14 GolbalPlanning::AstarCalcGInput构体 参考

Public 属性

- int **tmp**
- int **from**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.15 GolbalPlanning::AstarCalcGOutput构体 参考

Public 属性

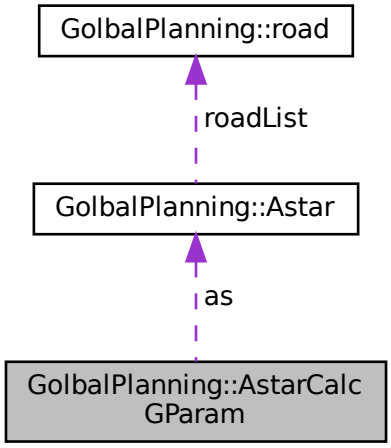
- double **result**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.16 GolbalPlanning::AstarCalcGParam构体 参考

GolbalPlanning::AstarCalcGParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.17 GolbalPlanning::AstarCalcHInput构体 参考

Public 属性

- int **tmp**
- int **destination**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.18 GolbalPlanning::AstarCalcHOutput构体 参考

Public 属性

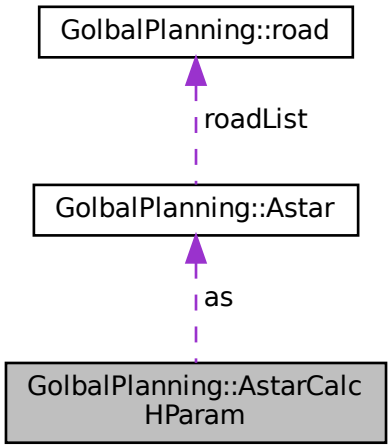
- double **result**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.19 GolbalPlanning::AstarCalcHParam构体 参考

GolbalPlanning::AstarCalcHParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.20 GolbalPlanning::AstarFindLaneInput构体 参考

Public 属性

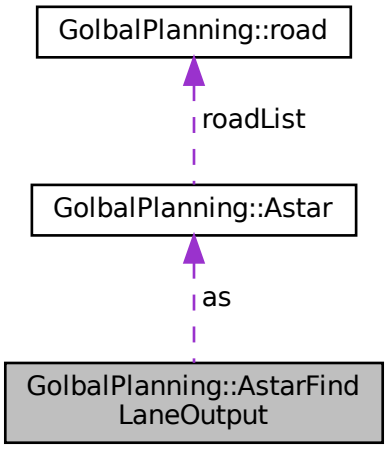
- RoadMap **m**
- list< int > **path**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.21 GolbalPlanning::AstarFindLaneOutput构体 参考

GolbalPlanning::AstarFindLaneOutput 的作:



Public 属性

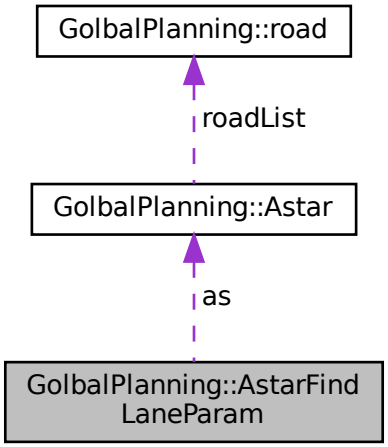
- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.22 GolbalPlanning::AstarFindLaneParam构体 参考

GolbalPlanning::AstarFindLaneParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.23 GolbalPlanning::AstarFindLaneRInput构体 参考

Public 属性

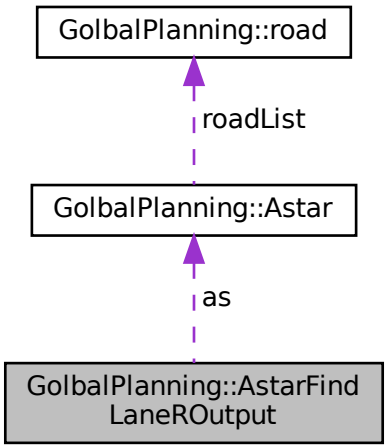
- vector< Road > **roads_**
- list< int > **path_**
- vector< std::pair< int, int > > **pathLanes_**
- int **lane_id_**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.24 GolbalPlanning::AstarFindLaneROutput构体 参考

GolbalPlanning::AstarFindLaneROutput 的作:



Public 属性

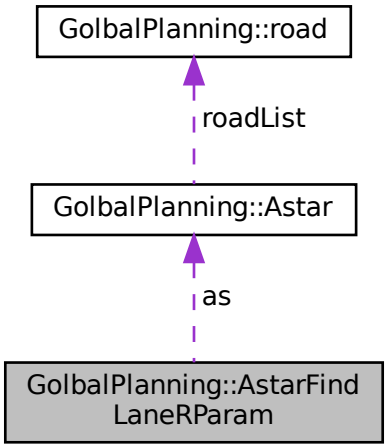
- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.25 GolbalPlanning::AstarFindLaneRParam构体 参考

GolbalPlanning::AstarFindLaneRParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.26 GolbalPlanning::AstarFindPathInput构体 参考

Public 属性

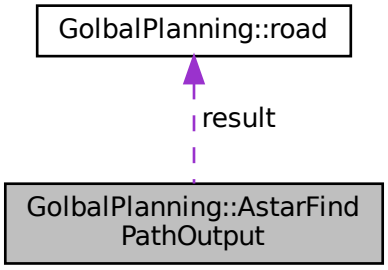
- int **origin**
- int **destination**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.27 GolbalPlanning::AstarFindPathOutput构体 参考

GolbalPlanning::AstarFindPathOutput 的作:



Public 属性

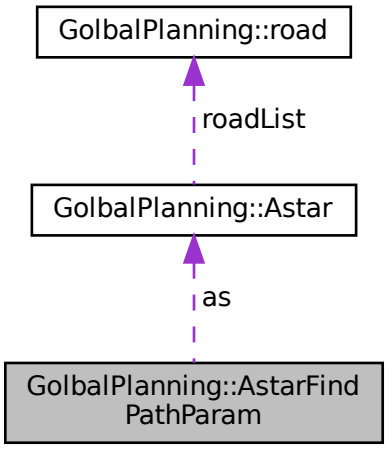
- road * **result**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.28 GolbalPlanning::AstarFindPathParam构体 参考

GolbalPlanning::AstarFindPathParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.29 GolbalPlanning::AstarGetLeastFInput构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.30 GolbalPlanning::AstarGetLeastFOutput构体 参考

Public 属性

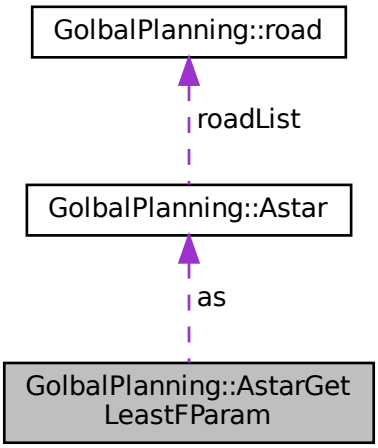
- int **result**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.31 GolbalPlanning::AstarGetLeastFParam构体 参考

GolbalPlanning::AstarGetLeastFParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.32 GolbalPlanning::AstarGetNextRoadInput构体 参考

Public 属性

- int **temp**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.33 GolbalPlanning::AstarGetNextRoadOutput构体 参考

Public 属性

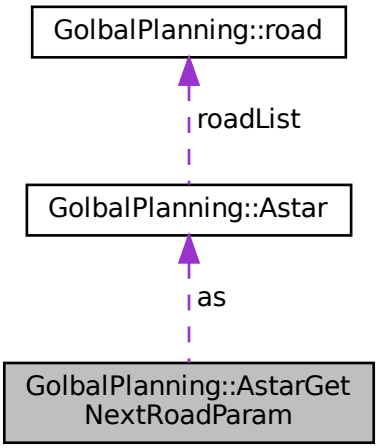
- std::vector< int > **result**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.34 GolbalPlanning::AstarGetNextRoadParam构体 参考

GolbalPlanning::AstarGetNextRoadParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.35 GolbalPlanning::AstarGetPathInput构体 参考

Public 属性

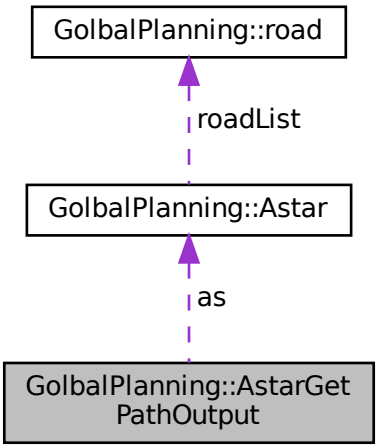
- int **origin**
- int **destination**
- int **originPointId**
- int **destinationPointId**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.36 GolbalPlanning::AstarGetPathOutput构体 参考

GolbalPlanning::AstarGetPathOutput 的作:



Public 属性

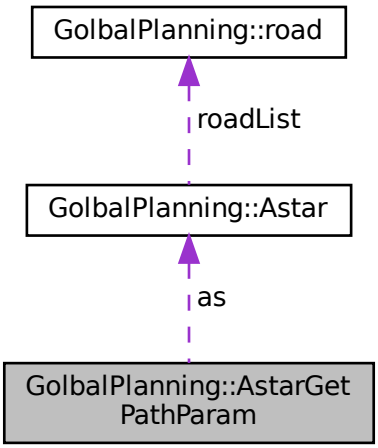
- std::list< int > **path**
- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.37 GolbalPlanning::AstarGetPathParam构体 参考

GolbalPlanning::AstarGetPathParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.38 GolbalPlanning::AstarInitLinkInput构体 参考

Public 属性

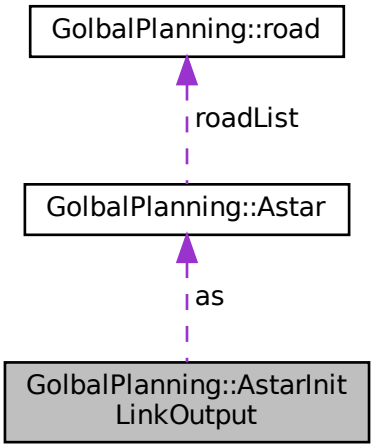
- int **from**
- int **to**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.39 GolbalPlanning::AstarInitLinkOutput构体 参考

GolbalPlanning::AstarInitLinkOutput 的作:



Public 属性

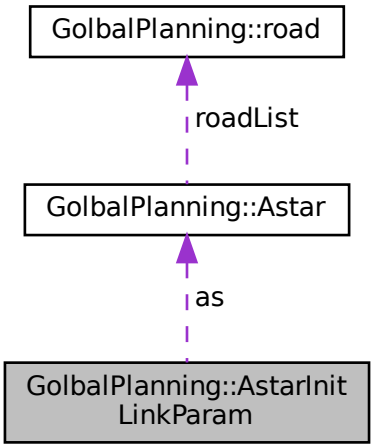
- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.40 GolbalPlanning::AstarInitLinkParam构体 参考

GolbalPlanning::AstarInitLinkParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.41 GolbalPlanning::AstarInitRoadInput构体 参考

Public 属性

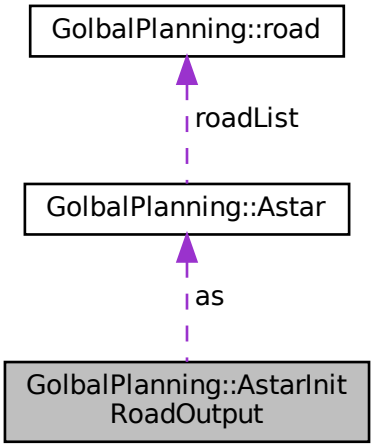
- int **number**
- double **xStart**
- double **yStart**
- double **xEnd**
- double **yEnd**
- double **length**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.42 GolbalPlanning::AstarInitRoadOutput构体 参考

GolbalPlanning::AstarInitRoadOutput 的作:



Public 属性

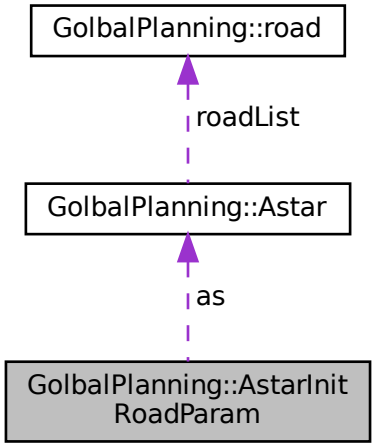
- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.43 GolbalPlanning::AstarInitRoadParam构体 参考

GolbalPlanning::AstarInitRoadParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

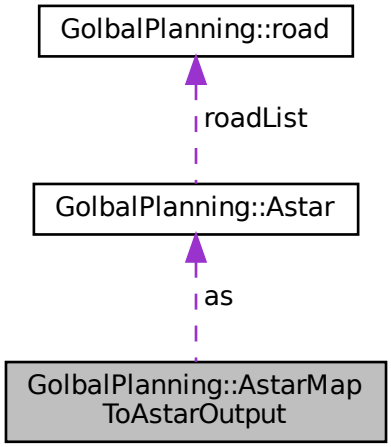
5.44 GolbalPlanning::AstarMapToAstarInput构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.45 GolbalPlanning::AstarMapToAstarOutput构体 参考

GolbalPlanning::AstarMapToAstarOutput 的作:



Public 属性

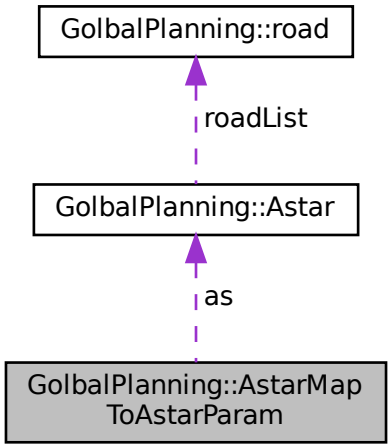
- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.46 GolbalPlanning::AstarMapToAstarParam构体 参考

GolbalPlanning::AstarMapToAstarParam 的作:



Public 属性

- RoadMap **m**
- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.47 GolbalPlanning::AstarModuleSelfCheckInput构体 参考

Public 属性

- std::list< int > **path**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

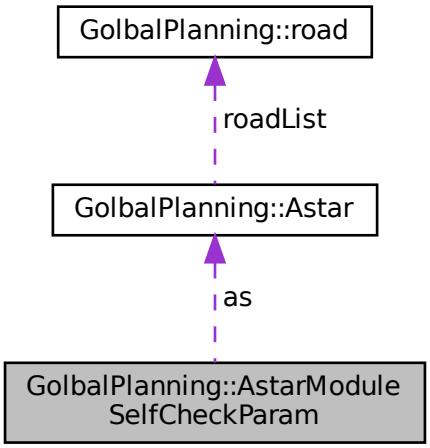
5.48 GolbalPlanning::AstarModuleSelfCheckOutput构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.49 GolbalPlanning::AstarModuleSelfCheckParam构体 参考

GolbalPlanning::AstarModuleSelfCheckParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.50 GolbalPlanning::AstarModuleSelfCheckPrintInput构体 参考

Public 属性

- vector< pair< int, int > > **pathLanes**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

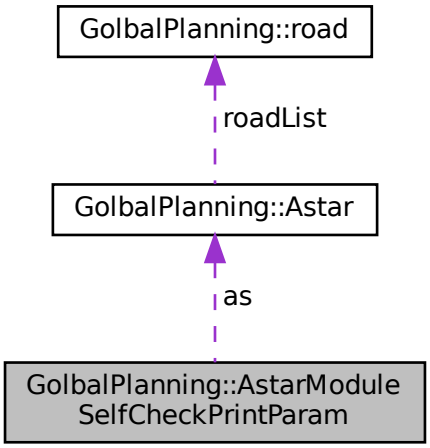
5.51 GolbalPlanning::AstarModuleSelfCheckPrintOutput构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.52 GolbalPlanning::AstarModuleSelfCheckPrintParam构体 参考

GolbalPlanning::AstarModuleSelfCheckPrintParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.53 GolbalPlanning::AstarRecursiveFindLaneInput构体 参考

Public 属性

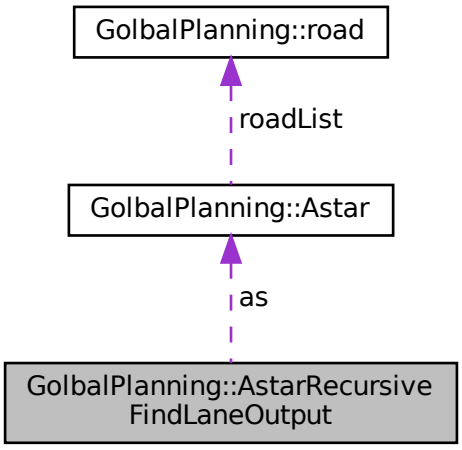
- std::vector< Road > **roads_**
- std::list< int > **path_**
- std::vector< std::tuple< int, int, int > > **optional_lanes_**
- int **road_id_**
- int **lane_id_**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.54 GolbalPlanning::AstarRecursiveFindLaneOutput构体 参考

GolbalPlanning::AstarRecursiveFindLaneOutput 的作:



Public 属性

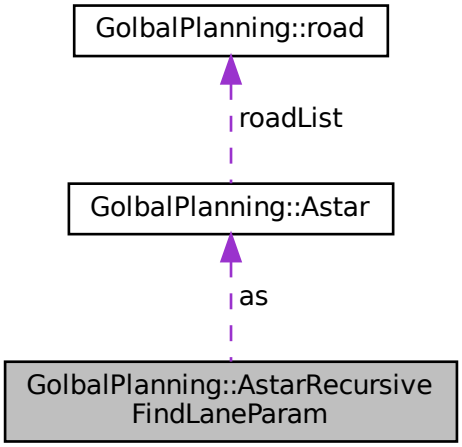
- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.55 GolbalPlanning::AstarRecursiveFindLaneParam构体 参考

GolbalPlanning::AstarRecursiveFindLaneParam 的作:



Public 属性

- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

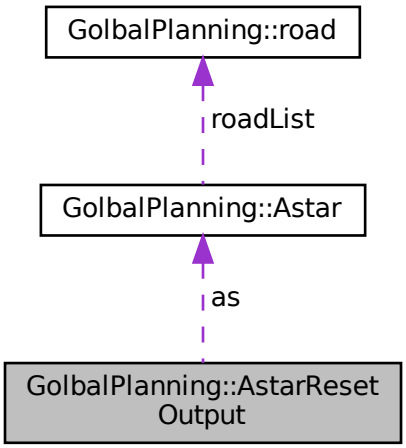
5.56 GolbalPlanning::AstarResetInput构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.57 GolbalPlanning::AstarResetOutput构体 参考

GolbalPlanning::AstarResetOutput 的作:



Public 属性

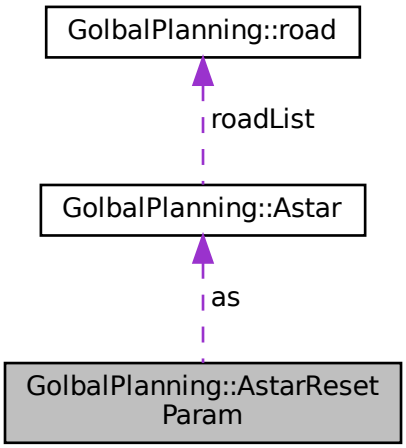
- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.58 GolbalPlanning::AstarResetParam构体 参考

GolbalPlanning::AstarResetParam 的作:



Public 属性

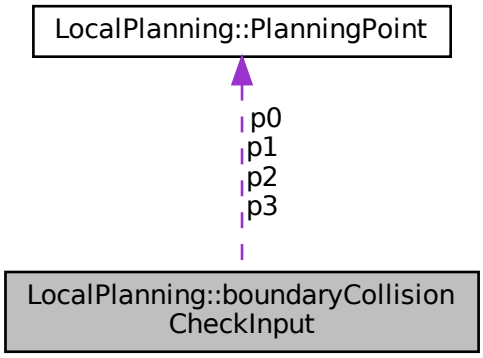
- Astar **as**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.59 LocalPlanning::boundaryCollisionCheckInput构体 参考

LocalPlanning::boundaryCollisionCheckInput 的作:



Public 属性

- PlanningPoint **p0**
- PlanningPoint **p1**
- PlanningPoint **p2**
- PlanningPoint **p3**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.60 LocalPlanning::boundaryCollisionCheckOutput构体 参考

Public 属性

- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

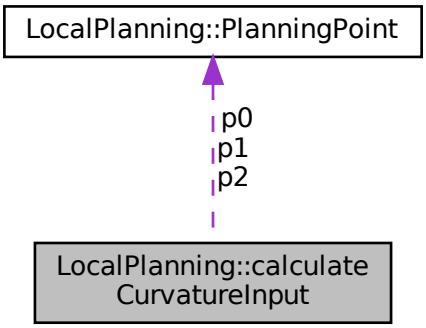
5.61 LocalPlanning::boundaryCollisionCheckParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.62 LocalPlanning::calculateCurvatureInput构体 参考

LocalPlanning::calculateCurvatureInput 的作:



Public 属性

- PlanningPoint **p0**
- PlanningPoint **p1**
- PlanningPoint **p2**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.63 LocalPlanning::calculateCurvatureOutput构体 参考

Public 属性

- double **curvature**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.64 LocalPlanning::calculateCurvatureParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.65 LocalPlanning::changeCurrentPointInput构体 参考

Public 属性

- int **laneId**
- int **pointIndex**
- GaussRoadPoint **currentPoint**
- bool **isLeft**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.66 LocalPlanning::changeCurrentPointOutput构体 参考

Public 属性

- int **laneId**
- int **pointIndex**
- GaussRoadPoint **currentPoint**
- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.67 LocalPlanning::changeCurrentPointParam构体 参考

Public 属性

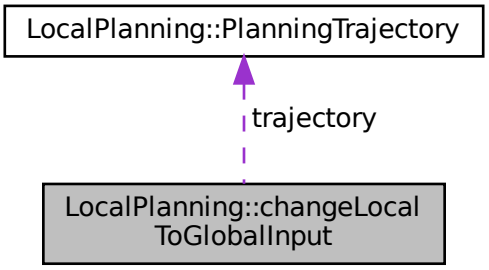
- Road **road**
- Lane **curLane**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.68 LocalPlanning::changeLocalToGlobalInput构体 参考

LocalPlanning::changeLocalToGlobalInput 的作:



Public 属性

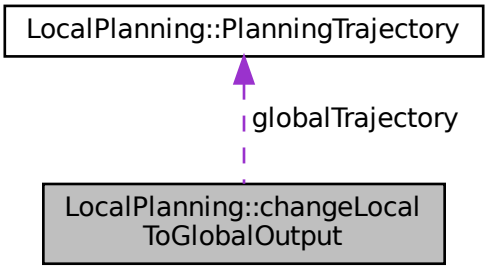
- PlanningTrajectory **trajectory**
- GaussRoadPoint **gaussRoadPoint**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.69 LocalPlanning::changeLocalToGlobalOutput构体 参考

LocalPlanning::changeLocalToGlobalOutput 的作:



Public 属性

- PlanningTrajectory **globalTrajectory**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.70 LocalPlanning::changeLocalToGlobalParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.71 LocalPlanning::clearPathListInput构体 参考

Public 属性

- std::vector< PlanningTrajectory > **pathList**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.72 LocalPlanning::clearPathListOutput构体 参考

Public 属性

- std::vector< PlanningTrajectory > **pathList**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.73 LocalPlanning::clearPathListParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.74 GolbalPlanning::cmp 参考

Public 成函数

- bool **operator()** (node a, node b) const

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.75 LocalPlanning::coordTran2DInput构体 参考

Public 属性

- double **dX**
- double **dY**
- double **dPhi**
- double **dX0**
- double **dY0**
- double **dPhi0**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.76 LocalPlanning::coordTran2DOutput构体 参考

Public 属性

- double **dX**
- double **dY**
- double **dPhi**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.77 LocalPlanning::coordTran2DParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.78 LocalPlanning::courseAngleReviseInput构体 参考

Public 属性

- double **ang**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.79 LocalPlanning::courseAngleReviseOutput构体 参考

Public 属性

- double **ang**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

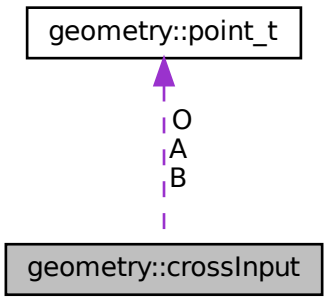
5.80 LocalPlanning::courseAngleReviseParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.81 geometry::crossInput构体 参考

geometry::crossInput 的作:



Public 属性

- point_t **O**
- point_t **A**
- point_t **B**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

5.82 geometry::crossOutput构体 参考

Public 属性

- double **value**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

5.83 geometry::crossParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

5.84 LocalPlanning::cruiseControllerInput构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩Planning_m.hpp

5.85 LocalPlanning::cruiseControllerOutput构体 参考

Public 属性

- double **velocity**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩Planning_m.hpp

5.86 LocalPlanning::cruiseControllerParam构体 参考

Public 属性

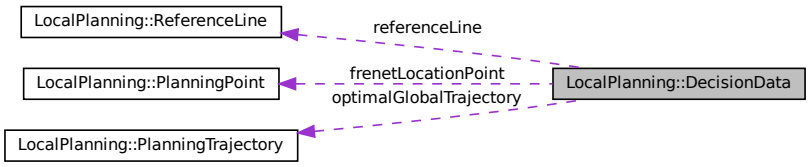
- pc::Imu **imu**
- std::vector< infopack::ObjectsProto > **objectsCmd**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.87 LocalPlanning::DecisionData构体 参考

LocalPlanning::DecisionData 的作:



Public 属性

- std::vector< PlanningTrajectory > **finalPathList**
- ReferenceLine **referenceLine**
- PlanningPoint **frenetLocationPoint**
- std::vector< std::vector< double > > **speedList**
- std::vector< PlanningTrajectory > **controlTrajectoryList**
- std::vector< int32_t > **feasibleTrajectoryIndexList**
- PlanningTrajectory **optimalGlobalTrajectory**
- int32_t **optimalTrajectoryIndex**
- int32_t **currentId** = 0
- int32_t **currentLaneId** = 0
- int32_t **currentIndex** = 0
- std::tuple< int32_t, int32_t > **nextId** = std::tuple<int32_t, int32_t>(0, 0)

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.88 LocalPlanning::findMaxInput构体 参考

Public 属性

- std::vector< double > **array**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.89 LocalPlanning::findMaxOutput构体 参考

Public 属性

- double **value**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.90 LocalPlanning::findMaxParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.91 LocalPlanning::findMinInput构体 参考

Public 属性

- std::vector< double > **array**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.92 LocalPlanning::findMinOutput构体 参考

Public 属性

- double **value**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.93 LocalPlanning::findMinParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.94 LocalPlanning::fixRoadLaneIndexInput构体 参考

LocalPlanning::fixRoadLaneIndexInput 的作:

Public 属性

- DecisionData **decisionData**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.95 LocalPlanning::fixRoadLaneIndexOutput构体 参考

Public 属性

- std::tuple< int32_t, int32_t > **indexTuple**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.96 LocalPlanning::fixRoadLaneIndexParam构体 参考

Public 属性

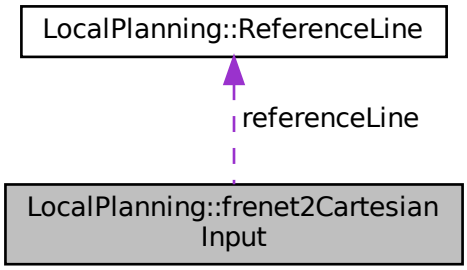
- RoadMap **map**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.97 LocalPlanning::frenet2CartesianInput构体 参考

LocalPlanning::frenet2CartesianInput 的作:



Public 属性

- double **s**
- double **l**
- double **x**
- double **y**
- ReferenceLine **referenceLine**
- int **lastIndex**
- int **prevIndex**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.98 LocalPlanning::frenet2CartesianOutput构体 参考

Public 属性

- double **x**
- double **y**
- int **lastIndex**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.99 LocalPlanning::frenet2CartesianParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.100 LocalPlanning::gaussConvertInput构体 参考

Public 属性

- double **longitude1**
- double **latitude1**
- double **dNorth_X**
- double **dEast_Y**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.101 LocalPlanning::gaussConvertOutput构体 参考

Public 属性

- double **dNorth_X**
- double **dEast_Y**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

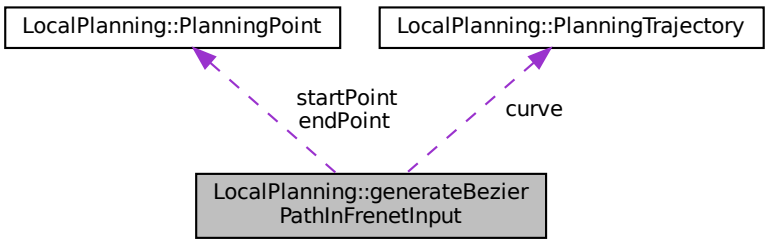
5.102 LocalPlanning::gaussConvertParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.103 LocalPlanning::generateBezierPathInFrenetInput构体 参考

LocalPlanning::generateBezierPathInFrenetInput 的作:



Public 属性

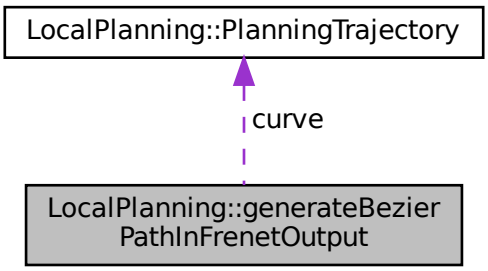
- PlanningPoint **startPoint**
- PlanningPoint **endPoint**
- PlanningTrajectory **curve**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.104 LocalPlanning::generateBezierPathInFrenetOutput构体 参考

LocalPlanning::generateBezierPathInFrenetOutput 的作:



Public 属性

- PlanningTrajectory **curve**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

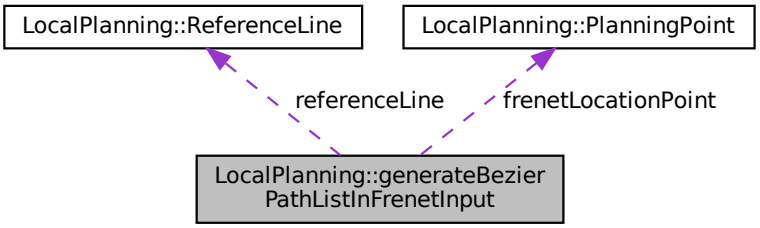
5.105 LocalPlanning::generateBezierPathInFrenetParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.106 LocalPlanning::generateBezierPathListInFrenetInput构体 参考

LocalPlanning::generateBezierPathListInFrenetInput 的作:



Public 属性

- ReferenceLine **referenceLine**
- PlanningPoint **frenetLocationPoint**
- std::vector< PlanningTrajectory > **pathList**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.107 LocalPlanning::generateBezierPathListInFrenetOutput构体 参考

Public 属性

- std::vector< PlanningTrajectory > **pathList**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

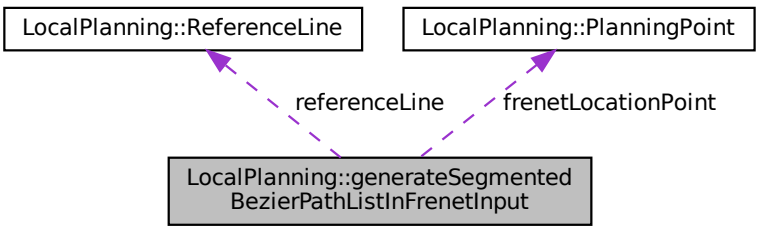
5.108 LocalPlanning::generateBezierPathListInFrenetParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.109 LocalPlanning::generateSegmentedBezierPathListInFrenetInput构体 参考

LocalPlanning::generateSegmentedBezierPathListInFrenetInput 的作:



Public 属性

- ReferenceLine **referenceLine**
- PlanningPoint **frenetLocationPoint**
- std::vector< PlanningTrajectory > **pathList**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.110 LocalPlanning::generateSegmentedBezierPathListInFrenetOutput 构体 参考

Public 属性

- std::vector< PlanningTrajectory > **pathList**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.111 LocalPlanning::generateSegmentedBezierPathListInFrenetParam 构体 参考

Public 属性

- RoadMap **map**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.112 LocalPlanning::generateSpeedListInput构体 参考

LocalPlanning::generateSpeedListInput 的作:

Public 属性

- double **velocity**
- DecisionData **decisionData**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.113 LocalPlanning::generateSpeedListOutput构体 参考

LocalPlanning::generateSpeedListOutput 的作:

Public 属性

- DecisionData **decisionData**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.114 LocalPlanning::generateSpeedListParam构体 参考

Public 属性

- std::vector< infopack::ObjectsProto > **objectsCmd**
- pc::Imu **imu**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.115 LocalPlanning::generateTrajectoryListInput构体 参考

LocalPlanning::generateTrajectoryListInput 的作:

Public 属性

- DecisionData **decisionData**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.116 LocalPlanning::generateTrajectoryListOutput构体 参考

LocalPlanning::generateTrajectoryListOutput 的作:

Public 属性

- DecisionData **decisionData**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.117 LocalPlanning::generateTrajectoryListParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.118 LocalPlanning::getAngleDiffInput构体 参考

Public 属性

- double **ang0**
- double **ang1**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.119 LocalPlanning::getAngleDiffOutput构体 参考

Public 属性

- double **ret**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.120 LocalPlanning::getAngleDiffParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.121 LocalPlanning::getAngleInput构体 参考

Public 属性

- double **x0**
- double **y0**
- double **x1**
- double **y1**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.122 LocalPlanning::getAngleOutput构体 参考

Public 属性

- double **ang**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.123 LocalPlanning::getAngleParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.124 LocalPlanning::getCurrentPositionInput构体 参考

LocalPlanning::getCurrentPositionInput 的作:

Public 属性

- DecisionData **decisionData**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.125 LocalPlanning::getCurrentPositionOutput构体 参考

LocalPlanning::getCurrentPositionOutput 的作:

Public 属性

- DecisionData **decisionData**
- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.126 LocalPlanning::getCurrentPositionParam构体 参考

Public 属性

- pc::Imu **imu**
- RoadMap **map**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.127 LocalPlanning::getDistanceInput构体 参考

Public 属性

- double **x1**
- double **y1**
- double **x2**
- double **y2**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.128 LocalPlanning::getDistanceOutput构体 参考

Public 属性

- double **dis**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

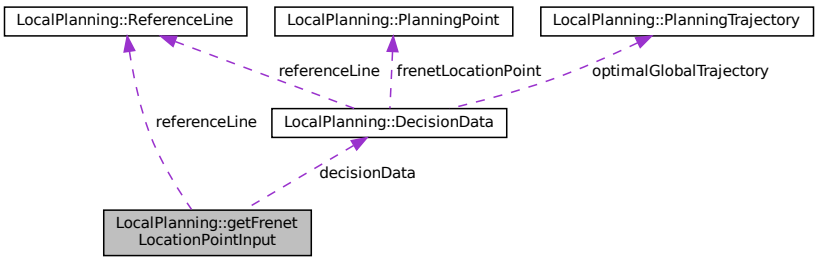
5.129 LocalPlanning::getDistanceParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.130 LocalPlanning::getFrenetLocationPointInput构体 参考

LocalPlanning::getFrenetLocationPointInput 的作:



Public 属性

- DecisionData **decisionData**
- ReferenceLine **referenceLine**
- GaussRoadPoint **locationPoint**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.131 LocalPlanning::getFrenetLocationPointOutput构体 参考

LocalPlanning::getFrenetLocationPointOutput 的作:

Public 属性

- DecisionData **decisionData**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

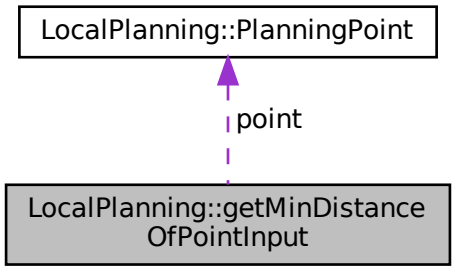
5.132 LocalPlanning::getFrenetLocationPointParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.133 LocalPlanning::getMinDistanceOfPointInput构体 参考

LocalPlanning::getMinDistanceOfPointInput 的作:



Public 属性

- PlanningPoint **point**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.134 LocalPlanning::getMinDistanceOfPointOutput构体 参考

Public 属性

- double **minDistance**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.135 LocalPlanning::getMinDistanceOfPointParam构体 参考

Public 属性

- prediction::ObjectList **prediction**
- pc::Imu **imu**
- std::vector< infopack::ObjectsProto > **objectsCmd**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.136 LocalPlanning::getNextRoadLaneIdInput构体 参考

Public 属性

- std::tuple< int32_t, int32_t > **currentId**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.137 LocalPlanning::getNextRoadLaneIdOutput构体 参考

Public 属性

- std::tuple< int32_t, int32_t > **nextRoadLaneId**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.138 LocalPlanning::getNextRoadLaneIdParam构体 参考

Public 属性

- std::vector< std::tuple< int32_t, int32_t > > **routingList**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.139 LocalPlanning::getOptimalTrajectoryIndexInput构体 参考

LocalPlanning::getOptimalTrajectoryIndexInput 的作:

Public 属性

- DecisionData **decisionData**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.140 LocalPlanning::getOptimalTrajectoryIndexOutput构体 参考

LocalPlanning::getOptimalTrajectoryIndexOutput 的作:

Public 属性

- DecisionData **decisionData**
- int32_t **index**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.141 LocalPlanning::getOptimalTrajectoryIndexParam构体 参考

Public 属性

- prediction::ObjectList **prediction**
- infopack::TrafficLight **trafficLight**
- pc::Imu **imu**
- std::vector< infopack::ObjectsProto > **objectsCmd**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.142 LocalPlanning::getPlanningDistanceInput构体 参考

Public 属性

- double **velocity**
- double **curvature**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.143 LocalPlanning::getPlanningDistanceOutput构体 参考

Public 属性

- double **distance**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.144 LocalPlanning::getPlanningDistanceParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.145 LocalPlanning::getPlanningPointInput构体 参考

LocalPlanning::getPlanningPointInput 的作:

Public 属性

- GaussRoadPoint **currentPoint**
- DecisionData **decisionData**
- double **distance**
- double **yaw**
- std::tuple< int32_t, int32_t > **nextId**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.146 LocalPlanning::getPlanningPointOutput构体 参考

Public 属性

- GaussRoadPoint **ret**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.147 LocalPlanning::getPlanningPointParam构体 参考

Public 属性

- RoadMap **map**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.148 LocalPlanning::getReferenceLineInput构体 参考

LocalPlanning::getReferenceLineInput 的作:

Public 属性

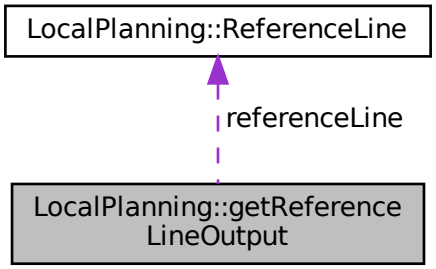
- DecisionData **decisionData**
- GaussRoadPoint **locationPoint**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.149 LocalPlanning::getReferenceLineOutput构体 参考

LocalPlanning::getReferenceLineOutput 的作:



Public 属性

- ReferenceLine **referenceLine**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.150 LocalPlanning::getReferenceLineParam构体 参考

Public 属性

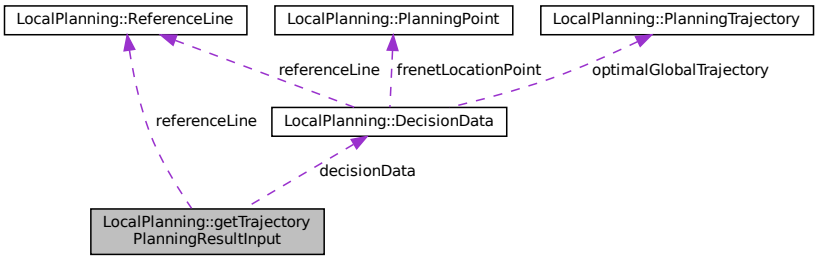
- RoadMap **map**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.151 LocalPlanning::getTrajectoryPlanningResultInput构体 参考

LocalPlanning::getTrajectoryPlanningResultInput 的作:



Public 属性

- double **velocity**
- DecisionData **decisionData**
- ReferenceLine **referenceLine**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.152 LocalPlanning::getTrajectoryPlanningResultOutput构体 参考

LocalPlanning::getTrajectoryPlanningResultOutput 的作:

Public 属性

- DecisionData **decisionData**
- int32_t **index**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.153 LocalPlanning::getTrajectoryPlanningResultParam构体 参考

Public 属性

- pc::Imu **imu**
- prediction::ObjectList **predictionMsg**
- RoadMap **map**
- infopack::TrafficLight **trafficLight**
- std::vector< infopack::ObjectsProto > **objectsCmd**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.154 LocalPlanning::hasPrecedingVehicleInput构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.155 LocalPlanning::hasPrecedingVehicleOutput构体 参考

Public 属性

- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.156 LocalPlanning::hasPrecedingVehicleParam构体 参考

Public 属性

- pc::Imu **imu**
- std::vector< infopack::ObjectsProto > **objectsCmd**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.157 LocalPlanning::inAreaInput构体 参考

Public 属性

- double **longitudeX**
- double **latitudeY**
- double **targetX**
- double **targetY**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.158 LocalPlanning::inAreaOutput构体 参考

Public 属性

- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.159 LocalPlanning::inAreaParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.160 LocalPlanning::initLocalPlanningInput构体 参考

LocalPlanning::initLocalPlanningInput 的作:

Public 属性

- DecisionData **decisionData**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.161 LocalPlanning::initLocalPlanningOutput构体 参考

LocalPlanning::initLocalPlanningOutput 的作:

Public 属性

- DecisionData **decisionData**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

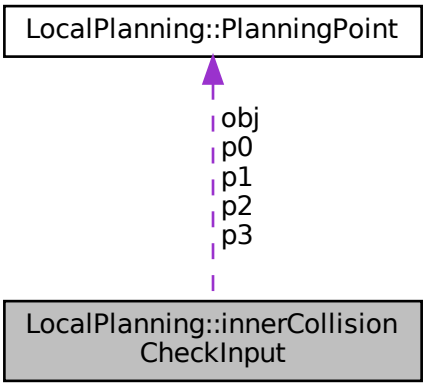
5.162 LocalPlanning::initLocalPlanningParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.163 LocalPlanning::innerCollisionCheckInput构体 参考

LocalPlanning::innerCollisionCheckInput 的作:



Public 属性

- PlanningPoint **p0**
- PlanningPoint **p1**
- PlanningPoint **p2**
- PlanningPoint **p3**
- PlanningPoint **obj**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.164 LocalPlanning::innerCollisionCheckOutput构体 参考

Public 属性

- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.165 LocalPlanning::innerCollisionCheckParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.166 geometry::inOrOutPolygonInput构体 参考

Public 属性

- int **nvert**
- double ∗ **vertx**
- double ∗ **verty**
- double **testx**
- double **testy**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩ _m.h

5.167 geometry::inOrOutPolygonOutput构体 参考

Public 属性

- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩ _m.h

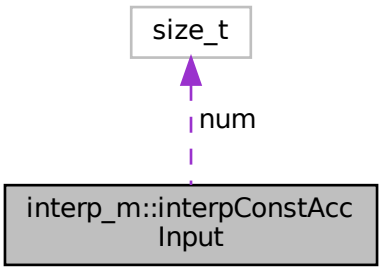
5.168 geometry::inOrOutPolygonParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩ _m.h

5.169 interp_m::interpConstAccInput构体 参考

interp_m::interpConstAccInput 的作:



Public 属性

- double **start_speed**
- double **end_speed**
- size_t **num**
- double **acceleration**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/interpolate↩_m.h

5.170 interp_m::interpConstAccOutput构体 参考

Public 属性

- std::vector< double > **result**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/interpolate↩_m.h

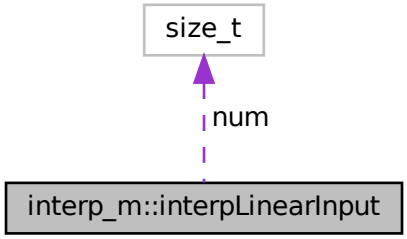
5.171 interp_m::interpConstAccParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/interpolate↩_m.h

5.172 interp_m::interpLinearInput构体 参考

interp_m::interpLinearInput 的作:



Public 属性

- double **start_speed**
- double **end_speed**
- size_t **num**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/interpolate↩_m.h

5.173 interp_m::interpLinearOutput构体 参考

Public 属性

- std::vector< double > **result**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/interpolate↩_m.h

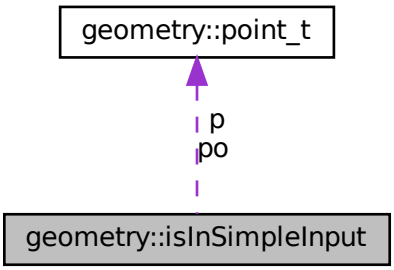
5.174 interp_m::interpLinearParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/interpolate↩_m.h

5.175 geometry::isInSimpleInput构体 参考

geometry::isInSimpleInput 的作:



Public 属性

- point_t * **p**
- int **n**
- point_t **po**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

5.176 geometry::isInSimpleOutput构体 参考

Public 属性

- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

5.177 geometry::isInSimpleParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

5.178 geometry::isLineIntersectionInput构体 参考

Public 属性

- GaussRoadPoint **line1StartPoint**
- GaussRoadPoint **line1EndPoint**
- GaussRoadPoint **line2StartPoint**
- GaussRoadPoint **line2EndPoint**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

5.179 geometry::isLineIntersectionOutput构体 参考

Public 属性

- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

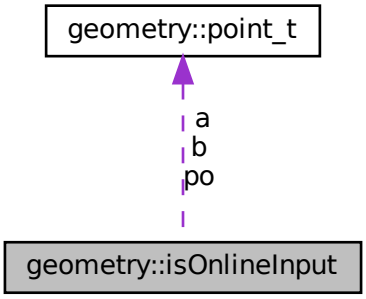
5.180 geometry::isLineIntersectionParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

5.181 geometry::isOnlineInput构体 参考

geometry::isOnlineInput 的作:



Public 属性

- point_t **a**
- point_t **b**
- point_t **po**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

5.182 geometry::isOnlineOutput构体 参考

Public 属性

- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

5.183 geometry::isOnlineParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

5.184 geometry::isPointInPolygonInput构体 参考

Public 属性

- std::vector< GaussRoadPoint > **pointPolygon**
- GaussRoadPoint **checkPoint**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

5.185 geometry::isPointInPolygonOutput构体 参考

Public 属性

- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

5.186 geometry::isPointInPolygonParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

5.187 geometry::isPolygonsIntersectionInput构体 参考

Public 属性

- std::vector< GaussRoadPoint > **polygon1Point**
- std::vector< GaussRoadPoint > **polygon2Point**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

5.188 geometry::isPolygonsIntersectionOutput构体 参考

Public 属性

- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

5.189 geometry::isPolygonsIntersectionParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

5.190 Jobs 参考

Public 成函数

- **Jobs** (std::vector< void * > &receivingSocketList, std::vector< void * > &sendingSocketList, RoadMap &m, std::vector< GaussRoadPoint > &stopPoints, std::tuple< int32_t, int32_t, int32_t > &stopPointRoadLaneId, std::vector< std::tuple< int32_t, int32_t >> &rList)
- virtual void **subscriber** ()
- virtual void **publisher** ()
- virtual void **processorLocalPlanning** ()
- virtual void **processorGlobalPlanning** ()
- virtual void **processorGlobalPlanningBaseLocal** ()
- virtual void **processorGlobalPlanningBaseServer** ()
- void **FindPointIDatRoad** (GaussRoadPoint roadPoint, int32_t roadID, int32_t &laneID, int32_t &pointID)
- virtual void **request** ()

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/planning.↩hpp
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/src/planning.cpp
- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/src/planningFigure_m.cpp

5.191 LocalPlanning::limitPlanningDistanceInput构体 参考

Public 属性

- double **targetDistance**
- double **lastTargetDistance**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩Planning_m.hpp

5.192 LocalPlanning::limitPlanningDistanceOutput构体 参考

Public 属性

- double **targetDistance**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.193 LocalPlanning::limitPlanningDistanceParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.194 LocalPlanning::loadStopPointsFromYamlInput构体 参考

Public 属性

- std::vector< GaussRoadPoint > **rawStopPoints**
- std::tuple< int32_t, int32_t, int32_t > **stopPointRoadLanePointId**
- double **offsetX**
- double **offsetY**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.195 LocalPlanning::loadStopPointsFromYamlOutput构体 参考

Public 属性

- std::vector< GaussRoadPoint > **rawStopPoints**
- std::tuple< int32_t, int32_t, int32_t > **stopPointRoadLanePointId**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.196 LocalPlanning::loadStopPointsFromYamlParam构体 参考

Public 属性

- RoadMap **map_**
- std::string **fileName**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.197 LocalPlanning::loadStopPointsInput构体 参考

Public 属性

- std::vector< GaussRoadPoint > **rawStopPoints**
- std::tuple< int32_t, int32_t, int32_t > **stopPointRoadLanePointId**
- double **offsetX**
- double **offsetY**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.198 LocalPlanning::loadStopPointsOutput构体 参考

Public 属性

- std::vector< GaussRoadPoint > **rawStopPoints**
- std::tuple< int32_t, int32_t, int32_t > **stopPointRoadLanePointId**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.199 LocalPlanning::loadStopPointsParam构体 参考

Public 属性

- std::string **fileName**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.200 GolbalPlanning::node 参考

Public 成函数

- **node** (int _number, double _F, int _father)

Public 属性

- int **number**
- double **F**
- int **father**

的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.201 LocalPlanning::PlanningPoint构体 参考

Public 属性

- double **x**
- double **y**
- double **angle**
- double **gaussX**
- double **gaussY**
- double **gaussAngle**
- double **s**
- double **l**
- double **frenetAngle**
- double **v**
- double **curvature**
- double **accumS**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.202 LocalPlanning::PlanningTrajectory构体 参考

Public 属性

- std::vector< PlanningPoint > **planningPoints**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.203 geometry::point_t构体 参考

Public 属性

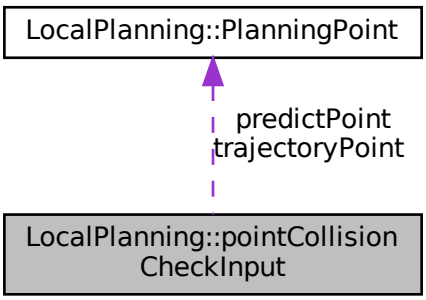
- double **x**
- double **y**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

5.204 LocalPlanning::pointCollisionCheckInput构体 参考

LocalPlanning::pointCollisionCheckInput 的作:



Public 属性

- PlanningPoint **trajectoryPoint**
- PlanningPoint **predictPoint**
- double **w**
- double **l**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩Planning_m.hpp

5.205 LocalPlanning::pointCollisionCheckOutput构体 参考

Public 属性

- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.206 LocalPlanning::pointCollisionCheckParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.207 LocalPlanning::pointDistanceInput构体 参考

Public 属性

- double **x1**
- double **x2**
- double **y1**
- double **y2**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.208 LocalPlanning::pointDistanceOutput构体 参考

Public 属性

- double **distance**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.209 LocalPlanning::pointDistanceParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.210 LocalPlanning::pointOnCubicBezierInput构体 参考

Public 属性

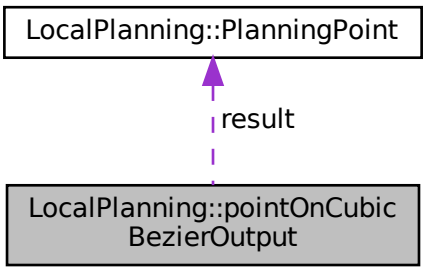
- std::vector< PlanningPoint > **cp**
- double **t**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.211 LocalPlanning::pointOnCubicBezierOutput构体 参考

LocalPlanning::pointOnCubicBezierOutput 的作:



Public 属性

- PlanningPoint **result**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.212 LocalPlanning::pointOnCubicBezierParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.213 LocalPlanning::pointToLineDistanceInput构体 参考

Public 属性

- GaussRoadPoint **startPoint**
- GaussRoadPoint **stopPoint**
- GaussRoadPoint **checkPoint**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.214 LocalPlanning::pointToLineDistanceOutput构体 参考

Public 属性

- double **dis**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

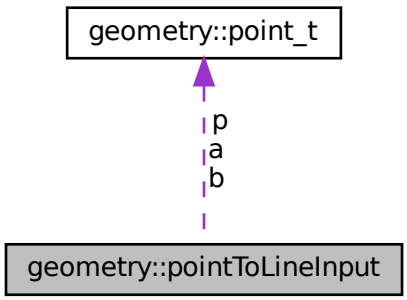
5.215 LocalPlanning::pointToLineDistanceParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.216 geometry::pointToLineInput构体 参考

geometry::pointToLineInput 的作:



Public 属性

- point_t **a**
- point_t **b**
- point_t **p**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

5.217 geometry::pointToLineOutput构体 参考

Public 属性

- double **value**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

5.218 geometry::pointToLineParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

5.219 geometry::pointToPolygonInput构体 参考

Public 属性

- std::vector< GaussRoadPoint > **pointPolygon**
- GaussRoadPoint **checkPoint**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

5.220 geometry::pointToPolygonOutput构体 参考

Public 属性

- double **dis**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

5.221 geometry::pointToPolygonParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/geometry↩_m.h

5.222 LocalPlanning::processTrafficLightInput构体 参考

LocalPlanning::processTrafficLightInput 的作:

Public 属性

- DecisionData **decisionData**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.223 LocalPlanning::processTrafficLightOutput构体 参考

LocalPlanning::processTrafficLightOutput 的作:

Public 属性

- DecisionData **decisionData**
- int **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.224 LocalPlanning::processTrafficLightParam构体 参考

Public 属性

- prediction::ObjectList **prediction**
- infopack::TrafficLight **trafficLight**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.225 LocalPlanning::ReferenceLine构体 参考

Public 属性

- std::vector< PlanningPoint > **referenceLinePoints**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.226 LocalPlanning::restartInput构体 参考

Public 属性

- double **velocity**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.227 LocalPlanning::restartOutput构体 参考

Public 属性

- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.228 LocalPlanning::restartParam构体 参考

Public 属性

- prediction::ObjectList **prediction**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.229 GolbalPlanning::road构体 参考

Public 属性

- int **number**
- double **xBegin**
- double **yBegin**
- double **xEnd**
- double **yEnd**
- double **length**
- std::vector< int > **to**
- int **father**
- int **isInList** = -2
- double **F**
- double **G**
- double **H**
- int **numLanes**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/global↩ Planning_m.h

5.230 LocalPlanning::stopInput构体 参考

LocalPlanning::stopInput 的作:

Public 属性

- DecisionData **decisionData**
- double **velocity**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.231 LocalPlanning::stopOutput构体 参考

LocalPlanning::stopOutput 的作:

Public 属性

- DecisionData **decisionData**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.232 LocalPlanning::stopParam构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.233 LocalPlanning::stopPointJudgeInput构体 参考

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.234 LocalPlanning::stopPointJudgeOutput构体 参考

Public 属性

- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.235 LocalPlanning::stopPointJudgeParam构体 参考

Public 属性

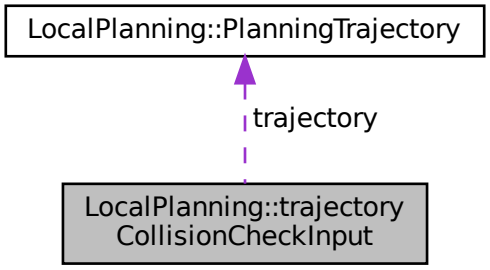
- pc::Imu **imu**
- std::vector< GaussRoadPoint > **stopPoints**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.236 LocalPlanning::trajectoryCollisionCheckInput构体 参考

LocalPlanning::trajectoryCollisionCheckInput 的作:



Public 属性

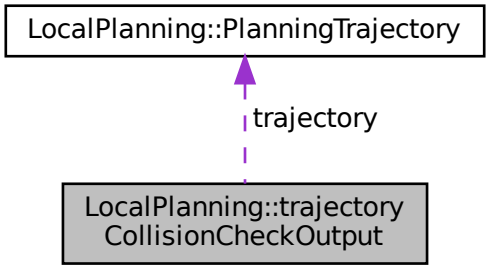
- PlanningTrajectory **trajectory**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.237 LocalPlanning::trajectoryCollisionCheckOutput构体 参考

LocalPlanning::trajectoryCollisionCheckOutput 的作:



Public 属性

- PlanningTrajectory **trajectory**
- bool **flag**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

5.238 LocalPlanning::trajectoryCollisionCheckParam构体 参考

Public 属性

- prediction::ObjectList **prediction**

构体的文档由以下文件生成:

- /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/include/local↩ Planning_m.hpp

Chapter 6

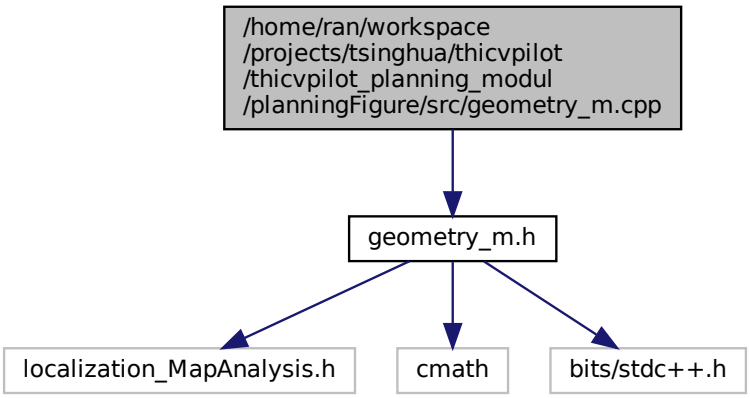
文件明

6.1 /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/src/geometry_m.cpp 文件参考

geometry

```
#include "geometry_m.h"
```

geometry_m.cpp 的引用(Include)系:



function

- geometry

  是一个算欧式距离的*namespace*。

函数

- void geometry::pointToPolygon (const pointToPolygonParam &param, const pointToPolygonInput &input, pointToPolygonOutput &output)

  算点到凸多形的距离

- void geometry::cross (const crossParam &param, const crossInput &input, crossOutput &output)

  点乘

- void geometry::pointToLine (const pointToLineParam &param, const pointToLineInput &input, pointToLine↩Output &output)

  算点到段的距离

- void geometry::isOnline (const isOnlineParam &param, const isOnlineInput &input, isOnlineOutput &output)

  判断点是否在段上

- void geometry::isInSimple (const isInSimpleParam &param, const isInSimpleInput &input, isInSimpleOutput &output)

  判断点是否在横平直的多形内部

- void geometry::isLineIntersection (const isLineIntersectionParam &param, const isLineIntersectionInput &input, isLineIntersectionOutput &output)

  判断两段是否相交

- void geometry::isPolygonsIntersection (const isPolygonsIntersectionParam &param, const isPolygons↩IntersectionInput &input, isPolygonsIntersectionOutput &output)

  判断两多形是否相交

- void geometry::isPointInPolygon (const isPointInPolygonParam &param, const isPointInPolygonInput &input, isPointInPolygonOutput &output)

  判断点是否在多形内部

- void geometry::inOrOutPolygon (const inOrOutPolygonParam &param, const inOrOutPolygonInput &input, inOrOutPolygonOutput &output)

  判断点是否在多形内部

作者

版本

1.0

日期

2024-12-04

版所有

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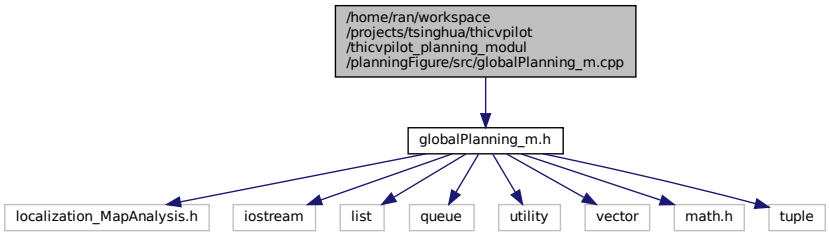
6.2 /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/src/globalPlanning_m.cpp 文件参考

globalPlanning

```
#include "globalPlanning_m.h"
```

globalPlanning_m.cpp 的引用(Include)系:



命名空

- GolbalPlanning

  是一个名*GolbalPlanning*的*namespace*。

函数

- void GolbalPlanning::AstarMapToAstar (const AstarMapToAstarParam &param, const AstarMapToAstarInput &input, AstarMapToAstarOutput &output)

  使用*Map*构建*Astar*
- void GolbalPlanning::AstarGetPath (const AstarGetPathParam &param, const AstarGetPathInput &input, AstarGetPathOutput &output)

  使用*Astar*算法取路径，包括理、后理等部分
- void GolbalPlanning::AstarInitRoad (const AstarInitRoadParam &param, const AstarInitRoadInput &input, AstarInitRoadOutput &output)

  *Astar*道路初始化
- void GolbalPlanning::AstarInitLink (const AstarInitLinkParam &param, const AstarInitLinkInput &input, AstarInitLinkOutput &output)

  *Astar*道路接系初始化
- void GolbalPlanning::AstarReset (const AstarResetParam &param, const AstarResetInput &input, Astar↩ResetOutput &output)

  在保留地的情况下，初始化*Astar*参数
- void GolbalPlanning::AstarModuleSelfCheck (const AstarModuleSelfCheckParam &param, const Astar↩ModuleSelfCheckInput &input, AstarModuleSelfCheckOutput &output)

  *Astar*道路*Id*
- void GolbalPlanning::AstarModuleSelfCheckPrint (const AstarModuleSelfCheckPrintParam &param, const AstarModuleSelfCheckPrintInput &input, AstarModuleSelfCheckPrintOutput &output)

  *Astar*道路*Id*打印
- void GolbalPlanning::AstarFindLane (const AstarFindLaneParam &param, const AstarFindLaneInput &input, AstarFindLaneOutput &output)

根据*Astar*道路全局划果，得道全局划果

- void GolbalPlanning::AstarFindLaneR (const AstarFindLaneRParam &param, const AstarFindLaneRInput &input, AstarFindLaneROutput &output)

  根据*Astar*道路全局划果，使用的方法得道全局划果

- void GolbalPlanning::AstarRecursiveFindLane (const AstarRecursiveFindLaneParam &param, const Astar↩ RecursiveFindLaneInput &input, AstarRecursiveFindLaneOutput &output)

  *Astar*得道全局划果主体函数

6.2.1 描述

globalPlanning

作者

版本

1.0

日期

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版所有

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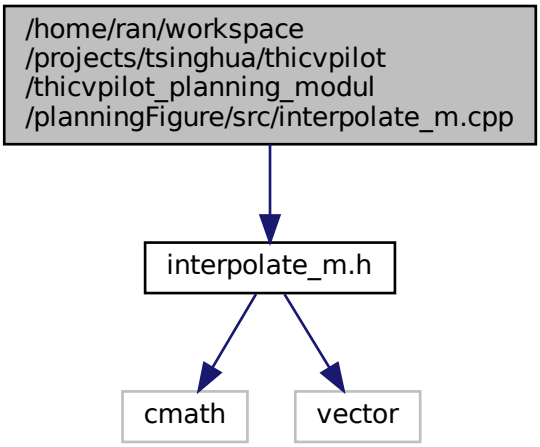
6.3 /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_↩ planning_modul/planningFigure/src/interpolate_m.cpp 文件参考

Interpolate speed along the path.

```
#include "interpolate_m.h"
```

interpolate_m.cpp 的引用(Include)系:



命名空

- interp_m

  是一个名*interp_m*的*namespace*。

函数

- void interp_m::linear (const interpLinearParam &param, const interpLinearInput &input, interpLinearOutput &output)

  根据初、和数，行性插
- void interp_m::constant_acceleration (const interpConstAccParam &param, const interpConstAccInput &input, interpConstAccOutput &output)

  在等隔的情况下，行匀加速度插

6.3.1 描述

Interpolate speed along the path.

作者

版本

1.0

日期

2024-12-04

版所有

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6.4 /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/src/localPlanning_m.cpp 文件参考

local planning

```
#include "localPlanning_m.hpp"
```

localPlanning_m.cpp 的引用(Include)系:

命名空

- LocalPlanning

  是一个名*LocalPlanning*的 *namespace*。

宏定

- #define **PLANNING_VELOCITY_MULTIPLIER** 2.6
- #define **DEFAULT_PLANNING_DISTANCE** 5
- #define **MAX_PLANNING_DISTANCE** 6
- #define **MIN_PLANNING_DISTANCE** 5
- #define **DISTANCE_RANGE** 3
- #define **DEFAULT_SECOND_PLANNING_DISTANCE_LONGER** 5
- #define **POINT_DIS_THRESHOLD_ON_CURRENT_LANE** 3
- #define **POINT_DIS_THRESHOLD_ON_OTHER_LANE** 2
- #define **POINT_ANGLE_THRESHOLD** 120
- #define **END_SPEED_NUM** 1
- #define **DESIRED_SPEED** 12 / 3.6
- #define **SPEED_RANGE** 3 / 3.6
- #define **CURVATURE_SPEED** 10 / 3.6
- #define **CURVATURE_THRESHOLS** 0.1
- #define **VEHICLE_DIAGONAL** 2.7951
- #define **VEHICLE_DIAGONAL_DEGREE** 26.5651
- #define **PREDICT_FREQUENCY** 0.1
- #define **STOP_JUDGE_DISTANCE** 6
- #define **MAX_FRENET_S** 8
- #define **ACC_RELA_DIST** 15.0
- #define **ACC_BEGIN_DIST** (ACC_RELA_DIST + 30)
- #define **ACC_FOLLOW_OBJECT_ID** 2
- #define **doNothing**()
- #define **SAFE_DISTANCE** 1.0

函数

- void LocalPlanning::gaussConvert (const gaussConvertParam &param, const gaussConvertInput &input, gaussConvertOutput &output)

  度全局坐
- void LocalPlanning::coordTran2D (const coordTran2DParam &param, const coordTran2DInput &input, coordTran2DOutput &output)

  全局坐局部坐
- void LocalPlanning::inArea (const inAreaParam &param, const inAreaInput &input, inAreaOutput &output)

  判断点是否在区域内
- void LocalPlanning::pointDistance (const pointDistanceParam &param, const pointDistanceInput &input, pointDistanceOutput &output)

  算两点之的距离
- void LocalPlanning::courseAngleRevise (const courseAngleReviseParam &param, const courseAngle↩ ReviseInput &input, courseAngleReviseOutput &output)

  逆旋坐系*90*度，向*0*度成向北*0*度
- void LocalPlanning::loadStopPoints (const loadStopPointsParam &param, const loadStopPointsInput &input, loadStopPointsOutput &output)

  从文件中加停止点
- void LocalPlanning::loadStopPointsFromYaml (const loadStopPointsFromYamlParam &param, const load↩ StopPointsFromYamlInput &input, loadStopPointsFromYamlOutput &output)

  从文件中加停止点
- void LocalPlanning::getNextRoadLaneId (const getNextRoadLaneIdParam &param, const getNextRoad↩ LaneIdInput &input, getNextRoadLaneIdOutput &output)

  从全局划中，取后道和道路*Id*
- void LocalPlanning::getPlanningDistance (const getPlanningDistanceParam &param, const getPlanning↩ DistanceInput &input, getPlanningDistanceOutput &output)

  根据速度、曲率取划距离
- void LocalPlanning::getPlanningPoint (const getPlanningPointParam &param, const getPlanningPointInput &input, getPlanningPointOutput &output)

  根据信息取划瞄点
- void LocalPlanning::limitPlanningDistance (const limitPlanningDistanceParam &param, const limitPlanning↩ DistanceInput &input, limitPlanningDistanceOutput &output)

  限制划距离，防止突
- void LocalPlanning::getAngle (const getAngleParam &param, const getAngleInput &input, getAngleOutput &output)

  算坐系中两点角度
- void LocalPlanning::getAngleDiff (const getAngleDiffParam &param, const getAngleDiffInput &input, get↩ AngleDiffOutput &output)

  算两角度之差，并正化
- void **LocalPlanning::localPlanning** (const pc::Imu &imu, double velocity, const RoadMap &map, Decision↩ Data &decisionData, const prediction::ObjectList &predictionMsg, const std::vector< std::tuple< int32_↩ t, int32_t >> &routingList, const std::vector< GaussRoadPoint > stopPoints, const infopack::TrafficLight &trafficLight, std::vector< infopack::ObjectsProto > objectsCmd)
- void LocalPlanning::restart (const restartParam &param, const restartInput &input, restartOutput &output)

  停后，判断能否重新后
- void LocalPlanning::stop (const stopParam &param, const stopInput &input, stopOutput &output)

  停修改速度曲
- void LocalPlanning::stopPointJudge (const stopPointJudgeParam &param, const stopPointJudgeInput &input, stopPointJudgeOutput &output)

  判断是否在停区域
- void LocalPlanning::pointOnCubicBezier (const pointOnCubicBezierParam &param, const pointOnCubic↩ BezierInput &input, pointOnCubicBezierOutput &output)

算塞曲上的点

- void LocalPlanning::generateBezierPathInFrenet (const generateBezierPathInFrenetParam &param, const generateBezierPathInFrenetInput &input, generateBezierPathInFrenetOutput &output)

  在*frenet*中生成塞曲

- void LocalPlanning::clearPathList (const clearPathListParam &param, const clearPathListInput &input, clearPathListOutput &output)

  清除上次生成的迹

- void LocalPlanning::getTrajectoryPlanningResult (const getTrajectoryPlanningResultParam &param, const getTrajectoryPlanningResultInput &input, getTrajectoryPlanningResultOutput &output)

  完成了多条迹和速度曲的生成，并化到笛卡坐系中，作出了迹的

- void LocalPlanning::getDistance (const getDistanceParam &param, const getDistanceInput &input, get↩DistanceOutput &output)

  算两点之的距离

- void LocalPlanning::pointToLineDistance (const pointToLineDistanceParam &param, const pointToLine↩DistanceInput &input, pointToLineDistanceOutput &output)

  算点到段的距离

- void LocalPlanning::angleRegulate (const angleRegulateParam &param, const angleRegulateInput &input, angleRegulateOutput &output)

  化角度到*0-180*中，便于比

- void LocalPlanning::getCurrentPosition (const getCurrentPositionParam &param, const getCurrentPosition↩Input &input, getCurrentPositionOutput &output)

  找距离位置最近的路点

- void LocalPlanning::generateSpeedList (const generateSpeedListParam &param, const generateSpeed↩ListInput &input, generateSpeedListOutput &output)

  根据道路信息等，采生成速度曲

- void LocalPlanning::generateTrajectoryList (const generateTrajectoryListParam &param, const generate↩TrajectoryListInput &input, generateTrajectoryListOutput &output)

  生成采迹，供后

- void LocalPlanning::processTrafficLight (const processTrafficLightParam &param, const processTrafficLight↩Input &input, processTrafficLightOutput &output)

  根据灯信息更新决策数据

- void LocalPlanning::getOptimalTrajectoryIndex (const getOptimalTrajectoryIndexParam &param, const get↩OptimalTrajectoryIndexInput &input, getOptimalTrajectoryIndexOutput &output)

  估迹，并取最迹索引

- void LocalPlanning::trajectoryCollisionCheck (const trajectoryCollisionCheckParam &param, const trajectoryCollisionCheckInput &input, trajectoryCollisionCheckOutput &output)

  迹碰撞，*true*表示可行，*false*表示碰撞

- void LocalPlanning::ableToPassYellowLight (const ableToPassYellowLightParam &param, const ableTo↩PassYellowLightInput &input, ableToPassYellowLightOutput &output)

  判断是否能通黄灯

- void LocalPlanning::pointCollisionCheck (const pointCollisionCheckParam &param, const pointCollision↩CheckInput &input, pointCollisionCheckOutput &output)

  迹点碰撞

- void LocalPlanning::boundaryCollisionCheck (const boundaryCollisionCheckParam &param, const boundaryCollisionCheckInput &input, boundaryCollisionCheckOutput &output)

  碰撞

- void LocalPlanning::innerCollisionCheck (const innerCollisionCheckParam &param, const innerCollision↩CheckInput &input, innerCollisionCheckOutput &output)

  内部包含

- void LocalPlanning::changeCurrentPoint (const changeCurrentPointParam &param, const changeCurrent↩PointInput &input, changeCurrentPointOutput &output)

  判断是否可以道

- void LocalPlanning::changeLocalToGlobal (const changeLocalToGlobalParam &param, const changeLocal↩ToGlobalInput &input, changeLocalToGlobalOutput &output)

  局部坐迹全局迹
- void LocalPlanning::initLocalPlanning (const initLocalPlanningParam &param, const initLocalPlanningInput &input, initLocalPlanningOutput &output)

  局部划决策数据初始化
- void LocalPlanning::fixRoadLaneIndex (const fixRoadLaneIndexParam &param, const fixRoadLaneIndex↩Input &input, fixRoadLaneIndexOutput &output)

  返回*decisionDate*中*roadID* 和*laneID*的索引
- void LocalPlanning::calculateCurvature (const calculateCurvatureParam &param, const calculateCurvature↩Input &input, calculateCurvatureOutput &output)

  由相三点近似算曲率
- void LocalPlanning::assessTrajectory (const assessTrajectoryParam &param, const assessTrajectoryInput &input, assessTrajectoryOutput &output)

  返回距离障碍物最近距离，行迹估
- void LocalPlanning::getMinDistanceOfPoint (const getMinDistanceOfPointParam &param, const getMin↩DistanceOfPointInput &input, getMinDistanceOfPointOutput &output)

  返回点距离障碍物最近距离
- void LocalPlanning::findMin (const findMinParam &param, const findMinInput &input, findMinOutput &output)

  返回*double*数最小
- void LocalPlanning::findMax (const findMaxParam &param, const findMaxInput &input, findMaxOutput &output)

  返回*double*数最大
- void LocalPlanning::getReferenceLine (const getReferenceLineParam &param, const getReferenceLineInput &input, getReferenceLineOutput &output)

  取局部划参考迹
- void LocalPlanning::getFrenetLocationPoint (const getFrenetLocationPointParam &param, const getFrenet↩LocationPointInput &input, getFrenetLocationPointOutput &output)

  取当前位置在*frenet*坐系中坐
- void LocalPlanning::frenet2Cartesian (const frenet2CartesianParam &param, const frenet2CartesianInput &input, frenet2CartesianOutput &output)

  将*frenet*坐系坐自坐系坐，*xy*右前
- void LocalPlanning::generateBezierPathListInFrenet (const generateBezierPathListInFrenetParam &param, const generateBezierPathListInFrenetInput &input, generateBezierPathListInFrenetOutput &output)

  在*frenet*坐系中生成多条塞曲
- void LocalPlanning::hasPrecedingVehicle (const hasPrecedingVehicleParam &param, const hasPreceding↩VehicleInput &input, hasPrecedingVehicleOutput &output)

  判断前方是否有需要跟随的
- void LocalPlanning::cruiseController (const cruiseControllerParam &param, const cruiseControllerInput &input, cruiseControllerOutput &output)

  *acc*景下，目速度算

6.4.1 描述

local planning

作者

版本

1.0

日期

2024-12-04

版所有

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6.5 /home/ran/workspace/projects/tsinghua/thicvpilot/thicvpilot_planning_modul/planningFigure/src/planningFigure_m.cpp 文件参考

planning

```
#include "PlanningMsg.pb.h"
#include "defineColor.h"
#include "globalPlanning_m.h"
#include "localPlanning_m.hpp"
#include "opencv2/highgui/highgui.hpp"
#include "opencv2/imgproc/imgproc.hpp"
#include "planning.hpp"
#include "sys/time.h"
#include "unistd.h"
#include <chrono>
#include <thread>
```

planningFigure_m.cpp 的引用(Include)系:

宏定

- #define **STOP_POINT_FILE_NAME** "../mapMsg/stopPoint.txt"
- #define **STOP_POINT_FILE_NAME_YAML** "../mapMsg/stopPoint.yaml"
- #define **RECV_PERIOD** 40
- #define **SEND_PERIOD** 100
- #define **GLOBAL_PLANNING_PERIOD** 100
- #define **LOCAL_PLANNING_PERIOD** 100
- #define **MAP_PATH** "../mapMsg/roadMap_KCYTemp.xodr"
- #define **gridInOneMeter** 5
- #define **mapSizeXMeter** 120
- #define **mapSizeYMeter** 120
- #define **mapYZero** 200
- #define **HEARTBEAT_PERIOD** 500
- #define **myselfVehicleID** 15
- #define **myselfVehicleeType** 0
- #define **myselfVCType** 1

函数

- cv::Point **drawRegulate** (cv::Point input)
- long int **cvtThreadId2Long** (thread::id id)
- void **checkThreadStatus** (int tStatus, thread::id id)
- int **main** ()

量

- auto **lastTheadTime** = std::chrono::steady_clock::now()
- auto **lastTheadTimeofSub** = std::chrono::steady_clock::now()
- auto **durationFroShowofSub** = std::chrono::duration_cast<std::chrono::milliseconds>(std::chrono↩ ::steady_clock::now() - lastTheadTimeofSub)

6.5.1 描述

planning

作者

版本

1.0

日期

2024-12-04

版所有

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